

Hyperbolic PDE-constrained decision-dependent robust planning of hydrogen energy storage systems for resilient electricity-water-gas nexus

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HIGHLIGHTS

- Multi-network synergies governed by hyperbolic partial differential equations (HPDEs) are exploited to mitigate multi-energy load shedding.
- Formulation of an HPDE-constrained optimal planning model of hydrogen energy storage systems for a resilient electricity–water–gas nexus.
- Development of a two-stage stochastic-robust planning strategy that jointly addresses exogenous uncertainty and endogenous uncertainty.
- Design of a tailored solution algorithm to reformulate the HPDE-constrained planning problem into a tractable MILP that is solved iteratively.
- Demonstration of significant improvements in system resilience and economic performance under power line outages.

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ABSTRACT

This paper proposes a decision-dependent robust planning method for hydrogen energy storage systems (HESSs) to enhance the resilience of the electricity-water-gas nexus (EWGN) under exogenous and endogenous uncertainties. In this method, the coordinated operations of power distribution networks, water distribution networks (WDNs), and gas distribution networks (GDNs), enhanced by HESSs, are incorporated to mitigate multi-energy load shedding. Moreover, hyperbolic partial differential equations (HPDEs) are employed to accurately represent the slow dynamics of water and gas pipeline flows in WDNs and GDNs, and then an HPDE-constrained HESS planning model is formulated to optimize the siting and sizing of HESSs. A two-stage stochastic-robust optimization strategy is developed to address exogenous and endogenous uncertainties so that the resulting HESS plan can hedge against the worst-case realization of decision-dependent hydrogen fuel cell vehicle refueling demand within each probabilistic exogenous scenario. Furthermore, a tailored solution algorithm, combining numerical discretization, piecewise linear relaxations, and parametric column-and-constraint generation, is devised to handle the proposed HPDE-constrained HESS planning model. Comparative studies show that the proposed method reduces the total load shedding cost by 81.66% compared with the case without HESS planning, and achieves a further 33.33% improvement in investment return over conventional planning methods without considering endogenous uncertainties.

1. Introduction

1.1. Motivation

The electricity-water-gas nexus (EWGN) has emerged as a critical infrastructure framework to address the growing interdependencies among power, water, and gas systems [1,2]. As a key component of

the EWGN, hydrogen energy storage systems (HESSs) have attracted growing interest for their ability to strengthen operational coordination across power distribution networks (PDNs), water distribution networks (WDNs), and gas distribution networks (GDNs) by producing hydrogen through water electrolysis and converting hydrogen to electricity using fuel cells [3–5]. Through this bidirectional energy conversion, HESSs can

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support critical electrical loads, sustain water pumping operations, and alleviate gas shortages to mitigate load shedding across PDNs, WDNs, and GDNs during network failures [6]. Given the increasing frequency of such failures, developing effective planning methods for HESSs has become essential to enhance the security, resilience, and survivability of the EWGN.

HESS planning necessarily involves the slow dynamics of WDNs and GDNs governed by hyperbolic partial differential equations (HPDEs) to accurately represent the physics of water and gas pipeline flows [7,8]. Moreover, HESS planning is further challenged by both exogenous and endogenous uncertainties. The former arises from stochastic renewable energy generation (REG), load fluctuations, and network failures [9,10]. The latter stems from the hydrogen refueling demand of hydrogen fuel cell vehicles (HFCVs), which is directly influenced by HESS planning decisions. Specifically, the locations and capacities of HESSs influence the accessibility and convenience of hydrogen refueling, which in turn affect drivers' refueling decisions and induce spatiotemporal variations in hydrogen refueling demand [11]. As a result, optimal HESS planning is an HPDE-constrained optimization problem under exogenous and endogenous uncertainties, which cannot be solved by conventional methods. Consequently, this paper aims to investigate a decision-dependent robust planning method for HESSs to enhance the EWGN's resilience.

1.2. Literature review

Driven by the growing integration of power and hydrogen systems in the EWGN, existing studies on hydrogen infrastructure planning have shifted from standalone hydrogen refueling stations [12] and power-to-gas facilities [13] to electricity-hydrogen integrated energy systems [14] and renewable-based hydrogen supply chains [15]. Recent studies have further emphasized the collaborative planning of integrated hydrogen energy chain multi-energy systems, where hydrogen production, compression, storage, transportation, and utilization are coordinated with other energy carriers [16]. For instance, a hydrogen hub-based planning framework was developed for power-gas-hydrogen integrated energy systems to optimize the location, size, and type of hydrogen-related components under demand and renewable uncertainties [17]. In addition, the portfolio selection and sizing of heterogeneous electrolyzer and fuel cell technologies in HESSs have been investigated for industrial park multi-energy systems, showing the importance of technology-level flexibility and hydrogen-water conversion efficiency in long-term planning [18]. More recently, hydrogen energy systems have also been investigated from a cross-sectoral chain coordination perspective, where hydrogen production, utilization, and refueling are jointly optimized to characterize the interactions among hydrogen production plants, hydrogen refueling stations, and HFCVs [19]. These efforts have primarily focused on electricity-hydrogen or power-gas-hydrogen coordination for optimal HESS siting and sizing to accommodate renewable variability and enhance energy flexibility. In response to growing hydrogen refueling demand driven by HFCVs, transportation networks have also been incorporated into hydrogen infrastructure planning to minimize HFCV users' travel costs [11,20]. However, most existing studies still emphasize economic or low-carbon objectives, with relatively limited attention to resilience-oriented HESS planning under network failures. Moreover, although recent research has considered the coordinated operation of power, gas, and hydrogen systems [17,21,22], the coordinated operation of PDNs, WDNs, and GDNs has not yet been fully incorporated into existing HESS planning studies. This limitation is nontrivial because HESSs simultaneously consume electricity and water for hydrogen production, provide hydrogen-based electricity generation, and inject synthetic natural gas into GDNs through methanation, thereby creating strong couplings among PDNs, WDNs, and GDNs.

The optimal planning model of HESSs must reflect the physics of water flows in WDNs and gas flows in GDNs to ensure reliable investment and operational decisions. To reduce modeling complexity, existing studies on hydrogen infrastructure planning have primarily adopted

steady-state gas flow models, such as the Weymouth equation [20,23]. These models capture temporal variations of flow and pressure only at pipeline terminals, neglecting the continuous spatiotemporal dynamics along pipelines [24]. In contrast to power flows, which reach a steady state within seconds, gas and water flows in pipelines exhibit slow dynamics, with transitions between steady states taking minutes to hours. Accurately representing this behavior requires dynamic modeling of water and gas flows governed by HPDEs [25]. Although dynamic gas flow models have been studied in operational contexts, they remain largely absent in HESS planning. A few studies have incorporated gas flow dynamics into the coordinated planning of power and hydrogen networks [7,24]. Nevertheless, they do not consider the dynamic behavior of water flows in WDNs, which is critical for capturing the impact of the time-varying water demand of HESSs on WDN operation. Therefore, an HESS planning model for the EWGN should simultaneously represent the slow dynamics of both water and gas pipeline flows, rather than relying only on steady-state or terminal-based network models.

Existing studies on hydrogen infrastructure planning commonly adopt stochastic optimization, robust optimization, and distributionally robust optimization to address uncertainties, such as those arising from REG, load fluctuations, and hydrogen refueling demand [7,9,13,20,26–29]. In these studies, uncertainties are typically treated as decision-independent uncertainties (DIUs), also known as exogenous uncertainties, which are assumed to remain unaffected by investment decisions. This assumption overlooks the demand-inducing effect, a well-documented phenomenon in transportation infrastructure planning where the deployment of refueling or charging stations stimulates additional user demand [30]. This introduces a form of uncertainty known as decision-dependent uncertainty (DDU), or endogenous uncertainty, which captures the mutual interactions between investment decisions and induced uncertain demand. Recent studies have begun to incorporate DDU into multi-energy infrastructure planning. For example, DDU has been modeled in integrated electricity and natural gas distribution system expansion planning by linking investment decisions with disaster-induced component fragility and resilience performance [31]. In hydrogen-related systems, a distributionally robust planning model with DDU was developed for hydrogen-electrical microgrids over sea islands, where infrastructure investment affects the ambiguity set of uncertain operating conditions [32]. For hydrogen refueling demand, Deng et al. [11] explored the link between hydrogen refueling station siting and HFCV adoption, while [14] incorporated induced refueling demand into networked hydrogen-electrical microgrid planning. However, the integration of DDUs into HESS planning under the coordinated operation of PDNs, WDNs, and GDNs remains unexplored. In particular, no existing work has jointly considered HESS siting and sizing, HPDE-constrained water and gas flow dynamics, exogenous uncertainties from renewable generation and network failures, and endogenous hydrogen refueling demand induced by HESS planning decisions.

1.3. Research contribution

In this paper, a decision-dependent robust planning method is proposed for HESSs to enhance the resilience of the EWGN considering both exogenous and endogenous uncertainties. The key contributions of this paper are threefold, as follows:

- 1) An HPDE-constrained optimal planning model is proposed for the siting and sizing of HESSs to enhance the resilience of the EWGN. In this model, HPDEs are employed to accurately represent the slow dynamics of water and gas pipeline flows in WDNs and GDNs. By incorporating the coordinated operations of PDNs, WDNs, and GDNs, enhanced by HESSs, the proposed model can exploit multi-network synergies to mitigate multi-energy load shedding.
- 2) A two-stage stochastic-robust optimization strategy is developed to address exogenous and endogenous uncertainties in HESS planning. A polyhedral DDU set is formulated to capture the

uncertainty in hydrogen refueling demand induced by HESS planning decisions, while exogenous DIUs are represented using a finite set of scenarios. The first stage determines HESS siting and sizing, and the second stage coordinates the operation of PDNs, WDNs, GDNs, and HESSs across all scenarios to minimize expected load shedding costs under the worst-case realization of endogenous uncertainty.

- 3) A tailored solution algorithm based on the parametric column-and-constraint generation (PC&CG) algorithm is devised to decompose the HESS planning problem into an investment master problem and multiple operation subproblems, enabling iterative convergence to an optimal planning solution. Numerical discretization combined with piecewise linear relaxation is employed to convert HPDE constraints into algebraic constraints, and then the HESS planning model can be reformulated into a tractable mixed-integer linear programming problem (MILP).

2. Coordinated multi-energy operation framework for enhancing EWGN resilience under network failures

Fig. 1 illustrates the coordinated multi-energy operation of the PDN, WDN, and GDN. The PDN integrates distributed renewable energy resources such as wind and photovoltaic generation together with substations and distribution feeders that supply electricity to a broad range of urban loads. The GDN is composed of compressors, pipelines, and gas turbines that support natural gas transmission and gas-to-power conversion. The WDN includes reservoirs, storage tanks, pumps, and pipelines that serve residential, public, and industrial water demands. These networks are linked through several well-known coupling mechanisms. Gas turbines convert natural gas into electricity and form a direct linkage between the PDN and the GDN. Water pumps depend on electricity for hydraulic transport and can also provide flexible demand that helps the PDN manage peak loads and integrate renewable generation. Beyond these traditional interactions, HESSs function as a central hub that strengthens the interdependence among the PDN, WDN, and GDN. A representative HESS incorporates an electrolyzer, hydrogen tank, methanation reactor, hydrogen refueling station, and solid oxide fuel cell [33]. The electrolyzer uses electricity from the PDN and water from the WDN to produce hydrogen. The hydrogen can be stored in hydrogen tanks or converted into synthetic natural gas through methanation and then injected into the GDN. Solid oxide fuel cells are capable of generating electricity using stored hydrogen and can therefore provide additional flexibility to the PDN. The hydrogen refueling station utilizes the produced hydrogen to satisfy the fueling needs of HFCVs. Through these multiple conversion and exchange pathways, HESSs enable strong energy linkages among the PDN, WDN, and GDN and provide an essential physical foundation for enhancing the EWGN resilience under network failures.

Line outages in PDNs occur much more frequently than network failures in WDNs or GDNs and tend to affect a larger portion of the service area. As a result, resilience-oriented planning for multi-energy networks commonly treats PDN line outages as the primary disturbance. Since WDNs and GDNs mainly operate through underground pipelines with relatively low failure probabilities and more localized failure impacts, this paper focuses on line outages in PDNs in the HESS planning of the EWGN. Under a line outage, HESSs provide critical support for multi-energy load restoration by enabling flexible energy conversion and cross-network support. Solid oxide fuel cells can utilize stored hydrogen to supply emergency power to priority electrical loads and sustain pump operation within the WDN, thereby reducing the likelihood of water service disruptions. At the same time, the loss of electrical capacity often requires gas turbines to increase their output to compensate for the power deficit. This response drives a sharp rise in natural gas consumption, which may create local pressure or flow constraints in the GDN. By producing synthetic natural gas via methanation and injecting it into the GDN, HESSs can mitigate these temporary supply stresses and

help stabilize GDN operation. Overall, during line outage events, HESSs serve as a key resource that strengthens the coordinated resilience of the PDN, WDN, and GDN. Their capability to transfer and reallocate energy among carriers reduces operational stress, limits multi-energy load shedding, and enhances the EWGN resilience.

3. HPDE-constrained HESS planning model formulation

3.1. Electricity-water-gas coupling modeling of HESS operation

Let $\mathcal{K}$ denote the set of candidate locations for deploying HESSs, let $\mathcal{T} := \{1, 2, \dots, T\}$ be the set of discrete operating periods with T being a finite time horizon, and let $t \in \mathcal{T}$ represent the time index, with Δt denoting the duration of each $t \in \mathcal{T}$.

The electrolyzer draws electricity from the PDN and water from the WDN to produce hydrogen, which can then be delivered to either the hydrogen storage tank or the methanation reactor. The mathematical model of the electrolyzer is given by Qiu et al. [34]:

$$\omega^h H_{k,t}^{\text{elz}} = \eta^{\text{elz}} P_{k,t}^{\text{elz}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (1a)$$

$$\eta^w H_{k,t}^{\text{elz}} = q_{k,t}^{\text{elz}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (1b)$$

$$H_{k,t}^{\text{elz}} = H_{k,t}^{\text{elz,ht}} + H_{k,t}^{\text{elz,mr}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (1c)$$

$$0 \leq P_{k,t}^{\text{elz}} \leq C_k^{\text{elz}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (1d)$$

where $P_{k,t}^{\text{elz}}$, $H_{k,t}^{\text{elz}}$, and $q_{k,t}^{\text{elz}}$ denote the electrolyzer power input, hydrogen production, and water consumption of the HESS located at location k and time t , respectively; $H_{k,t}^{\text{elz,ht}}$ and $H_{k,t}^{\text{elz,mr}}$ represent the portions of hydrogen delivered to the hydrogen storage tank and the methanation reactor; ω^h is the lower heating value of hydrogen; η^{elz} and η^w denote the electrolyzer efficiency and the water-to-hydrogen conversion efficiency; and C_k^{elz} represents the installed capacity of the electrolyzer at location k .

The hydrogen tank supplies hydrogen to the solid oxide fuel cell, the methanation reactor, and the hydrogen refueling station according to system operating requirements. Its mathematical model is given by:

$$S_{k,t+1}^{\text{ht}} = S_{k,t}^{\text{ht}} + \left(H_{k,t}^{\text{elz,ht}} \eta^{\text{ht,ch}} - \frac{H_{k,t}^{\text{ht,dis}}}{\eta^{\text{ht,dis}}} \right) \Delta t, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (2a)$$

$$0 \leq H_{k,t}^{\text{elz,ht}} \leq \gamma^{\text{ch}} C_k^{\text{ht}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (2b)$$

$$0 \leq H_{k,t}^{\text{ht,dis}} \leq \gamma^{\text{dis}} C_k^{\text{ht}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (2c)$$

$$H_{k,t}^{\text{ht,dis}} = H_{k,t}^{\text{ht,mr}} + H_{k,t}^{\text{fc}} + H_{k,t}^{\text{hrs}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (2d)$$

$$0 \leq S_{k,t}^{\text{ht}} \leq C_k^{\text{ht}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (2e)$$

where $S_{k,t}^{\text{ht}}$ and $H_{k,t}^{\text{ht,dis}}$ denote the hydrogen inventory and the hydrogen discharge of the hydrogen tank in the HESS at location k and time t , respectively; $\eta^{\text{ht,ch}}$ and $\eta^{\text{ht,dis}}$ represent the charging and discharging efficiencies of the hydrogen tank; $H_{k,t}^{\text{ht,mr}}$, $H_{k,t}^{\text{fc}}$, and $H_{k,t}^{\text{hrs}}$ denote the hydrogen flows delivered from the hydrogen tank to the methanation reactor, the solid oxide fuel cell, and the hydrogen refueling station, respectively; C_k^{ht} denotes the installed capacity of the hydrogen tank in the HESS at location k ; and γ^{ch} and γ^{dis} specify the maximum charging and discharging rates as fractions of the installed capacity.

The methanation reactor converts hydrogen supplied by the electrolyzer and the hydrogen storage tank into synthetic natural gas through a catalytic reaction with carbon dioxide, and the produced gas is injected into the natural gas network. Its mathematical model is given by Qiu et al. [34]:

$$f_{k,t}^s = \eta^{\text{mr}} \frac{\omega^h (H_{k,t}^{\text{elz,mr}} + H_{k,t}^{\text{ht,mr}})}{\omega^{\text{ng}}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (3a)$$

$$0 \leq f_{k,t}^s \leq C_k^{\text{mr}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (3b)$$

where $f_{k,t}^s$ denotes the amount of synthetic natural gas produced by the methanation reactor in the HESS at location k and time t and injected

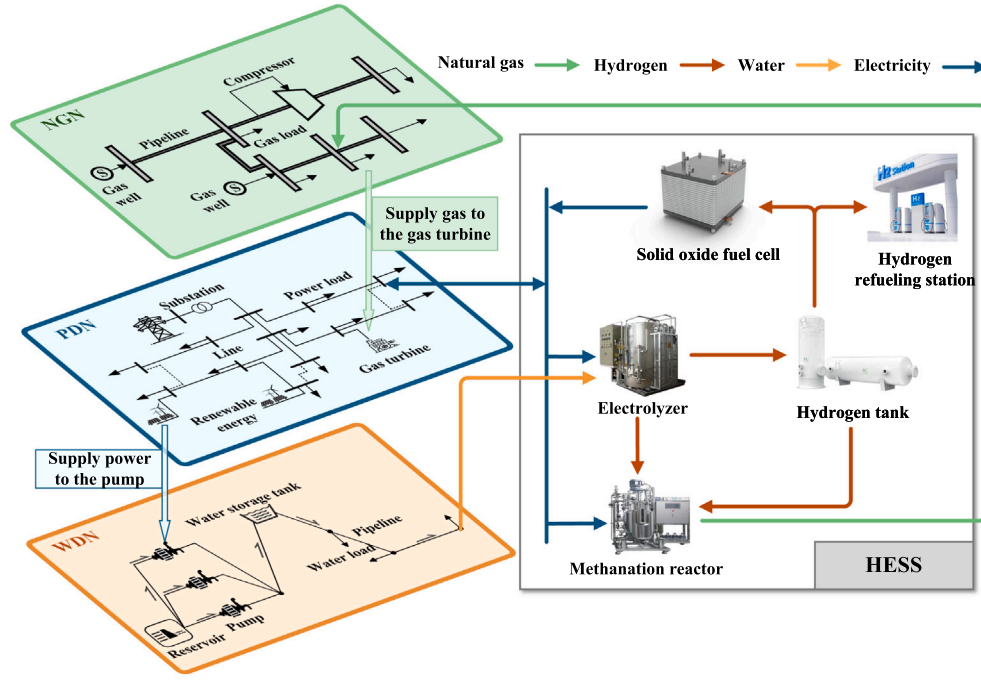


Fig. 1. Coordinated operation of the PDN, WDN, and GDN with HESSs.

into the GDN; η^{mr} represents the conversion efficiency of the methanation reactor; ω^{ng} is the lower heating value of natural gas; and C_k^{mr} denotes the installed capacity of the methanation reactor at location k .

The solid oxide fuel cell converts hydrogen into electricity and provides power support to the PDN during line outages. Its mathematical model is given by:

$$P_{k,t}^{\text{fc}} = \eta^{\text{fc,h}} H_{k,t}^{\text{fc}} \omega^{\text{h}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (4a)$$

$$0 \leq P_{k,t}^{\text{fc}} \leq C_k^{\text{fc}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (4b)$$

where $P_{k,t}^{\text{fc}}$ denotes the electrical output of the solid oxide fuel cell in the HESS at location k and time t ; $\eta^{\text{fc,h}}$ represents the conversion efficiency of hydrogen to electricity; and C_k^{fc} is the installed capacity of the solid oxide fuel cell at location k .

The hydrogen refueling station is the hydrogen supply unit within an HESS and serves HFCVs. Its mathematical model is given by:

$$H_{k,t}^{\text{hrs}} = \sigma_{\text{mv}}(d_{k,t} - d_{k,t}^{\text{sh}}), \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (5a)$$

$$\sum_{k \in \mathcal{K}} H_{k,t}^{\text{hrs}} = \sigma_{\text{mv}} \left(\sum_{k \in \mathcal{K}} d_{k,t} - \sum_{k \in \mathcal{K}} d_{k,t}^{\text{sh}} \right), \quad \forall t \in \mathcal{T}, \quad (5b)$$

$$0 \leq d_{k,t}^{\text{sh}} \leq d_{k,t}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (5c)$$

$$0 \leq H_{k,t}^{\text{hrs}} \leq C_k^{\text{hrs}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (5d)$$

where $d_{k,t}$ denotes the hydrogen demand of HFCVs served by the hydrogen refueling station in the HESS at location k and time t ; $d_{k,t}^{\text{sh}}$ represents the unserved portion of this demand; C_k^{hrs} denotes the installed capacity of the hydrogen refueling station at location k ; and σ_{mv} is the dimension conversion factor.

In addition, the internal power balance of each HESS must be satisfied and is expressed as follows:

$$P_{k,t}^{\text{fc}} + P_{k,t}^{\text{b,e}} = P_{k,t}^{\text{s,e}} + (1 + \lambda^{\text{cp}}) P_{k,t}^{\text{elz}} + \lambda^{\text{mr}} f_{k,t}^{\text{s}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (6a)$$

$$0 \leq P_{k,t}^{\text{b,e}} \leq \delta_{k,t}^{\text{b,e}} \bar{P}^{\text{b,e}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (6b)$$

$$0 \leq P_{k,t}^{\text{s,e}} \leq \delta_{k,t}^{\text{s,e}} \bar{P}^{\text{s,e}}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (6c)$$

$$\delta_{k,t}^{\text{s,e}} + \delta_{k,t}^{\text{b,e}} \leq 1, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \quad (6d)$$

where $P_{k,t}^{\text{b,e}}$ and $P_{k,t}^{\text{s,e}}$ denote the electricity imported from and exported to the PDN by the HESS at location k during the time period t , respectively; their upper bounds are given by $\bar{P}^{\text{b,e}}$ and $\bar{P}^{\text{s,e}}$; $\delta_{k,t}^{\text{s,e}}$ and $\delta_{k,t}^{\text{b,e}}$ are binary variables indicating the operating mode of the HESS, with $\delta_{k,t}^{\text{b,e}} = 1$ and $\delta_{k,t}^{\text{s,e}} = 0$ when electricity is imported from the PDN, and $\delta_{k,t}^{\text{s,e}} = 1$ and $\delta_{k,t}^{\text{b,e}} = 0$ when electricity is exported to the PDN; λ^{cp} represents the electricity consumption coefficient of the compressor in the electrolyzer; and λ^{mr} denotes the energy consumption coefficient of the methanation reactor.

3.2. HPDE-constrained dynamic energy flow modeling of the EWGN

The PDN, WDN, and GDN exhibit different dynamic characteristics. The PDN is dominated by fast system dynamics, such as electromagnetic and electromechanical transients, which usually settle well before the scheduling interval. Therefore, the post-contingency PDN state can be represented by an algebraic steady-state model [25]. In contrast, the WDN and GDN are governed by slow system dynamics arising from transport inertia and pressure/head propagation delays in pipelines, and their transition processes may last from tens of minutes to several hours, depending on network scale and operating conditions [8]. These slow system dynamics are important for HESS planning, since electrolyzers, methanation reactors, solid oxide fuel cells, and gas turbines induce time-varying water withdrawals and gas injections/consumptions. A steady-state treatment of the WDN and GDN would overestimate the immediacy of cross-network support and may bias the siting, sizing, and operation of HESSs. Therefore, PDE-based dynamic models are adopted for the WDN and GDN, while a steady-state model is used for the PDN.

The topologies of the PDN, WDN, and GDN can be described by connected directed graphs $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ are the sets of nodes and edges, respectively. Each edge between nodes $i, j \in \mathcal{V}$ is denoted by (i, j) . In this paper, superscripts e, w, and g are used to distinguish the graph components associated with the PDN, WDN, and GDN, respectively. Let $\mathcal{E}_{\text{rcs}}^{\text{e}} \subseteq \mathcal{E}^{\text{e}}$ be the set of lines equipped with a remote-control switch. For the GDN, $\mathcal{V}^{\text{g,well}} \subseteq \mathcal{V}^{\text{g}}$ and $\mathcal{V}^{\text{g,gt}} \subseteq \mathcal{V}^{\text{g}}$ denote the

sets of nodes connected to gas sources and gas turbines, respectively. Moreover, the edge set $\mathcal{E}^g \subseteq \mathcal{V}^g \times \mathcal{V}^g$ is composed of the gas pipeline set $\mathcal{E}^{g,\text{pip}}$ and the compressor set $\mathcal{E}^{g,\text{com}}$, i.e., $\mathcal{E}^g = \mathcal{E}^{g,\text{pip}} \cup \mathcal{E}^{g,\text{com}}$. For the WDN, $\mathcal{V}^{w,\text{rer}} \subseteq \mathcal{V}^w$ and $\mathcal{V}^{w,\text{tank}} \subseteq \mathcal{V}^w$ denote the sets of nodes connected to reservoirs and storage tanks, respectively. The edge set $\mathcal{E}^w \subseteq \mathcal{V}^w \times \mathcal{V}^w$ includes the water pipeline set $\mathcal{E}^{w,\text{pip}}$ and the pump set $\mathcal{E}^{w,\text{pump}}$, such that $\mathcal{E}^w = \mathcal{E}^{w,\text{pip}} \cup \mathcal{E}^{w,\text{pump}}$.

It is widely accepted in the literature that the following baseline assumptions apply to gas pipeline flow modeling [25]: gas pressure and density satisfy the ideal gas equation of state; the flow is adiabatic and isothermal; all pipelines are horizontal and have uniform diameter and internal surface roughness; the flow is turbulent with a high Reynolds number; and the flow direction in each pipeline is fixed and unidirectional. There is a well-established consensus that compressible gas flow in each pipeline $(i, j) \in \mathcal{E}^{g,\text{pip}}$ is well described by the one-dimensional isothermal Euler equations [35], which constitute a system of HPDEs:

$$A_{ij} \frac{\partial p_{ij}}{\partial t} + a^2 \frac{\partial f_{ij}}{\partial x} = 0, \quad (7a)$$

$$\frac{\partial f_{ij}}{\partial t} + A_{ij} \frac{\partial p_{ij}}{\partial x} + \frac{a^2}{A_{ij}} \frac{\partial}{\partial x} \left(\frac{f_{ij}^2}{p_{ij}} \right) = -\frac{\lambda a^2 f_{ij}^2}{2D_{ij} A_{ij} p_{ij}}, \quad (7b)$$

where the state variables f_{ij} and p_{ij} represent the mass flow rate and pressure in pipeline (i, j) , respectively. These variables depend on time $t \in [0, T]$ and space $x \in [0, L_{ij}]$, where L_{ij} denotes the length of pipeline (i, j) . Here, λ denotes the friction factor, D_{ij} is the internal diameter, A_{ij} is the cross-sectional area of pipeline (i, j) , and a is the pressure wave speed. Under the above assumptions, a , A_{ij} , D_{ij} , and λ can be treated as constants [25]. It is assumed that the initial conditions are given and that the admissible boundary conditions are specified according to the flow characteristics.

To describe the pressure relationships among different types of nodes and edges in a unified manner, the variable $p_{i,t}$ is introduced to denote the gas pressure at node $i \in \mathcal{V}^g$ at time t . Compressors are installed to compensate for pressure losses during natural gas transmission. It is assumed that the energy required for compressor operation is supplied by the natural gas flowing through the compressor itself. Since the gas consumption of a compressor is typically small, usually accounting for about 3%–5% of the transported gas, it is approximated as a constant proportion [8,36]. Let a_{ij}^{com} denote the gas consumption ratio of compressor $(i, j) \in \mathcal{E}^{g,\text{com}}$. Then, the relationship between the outlet and inlet gas flows, together with the compression-ratio limits, can be expressed as [23]:

$$f_{ij,t}^{\text{com,out}} = (1 - a_{ij}^{\text{com}}) f_{ij,t}^{\text{com,in}}, \quad \forall (i, j) \in \mathcal{E}^{g,\text{com}}, t \in [0, T], \quad (8a)$$

$$r_{ij,t}^{\text{com}} p_{i,t} \leq p_{j,t} \leq \bar{r}_{ij,t}^{\text{com}} p_{i,t}, \quad \forall (i, j) \in \mathcal{E}^{g,\text{com}}, t \in [0, T], \quad (8b)$$

where $f_{ij,t}^{\text{com,in}}$ and $f_{ij,t}^{\text{com,out}}$ denote the inlet and outlet gas flows of compressor (i, j) at time t , respectively, and $r_{ij,t}^{\text{com}}$ and $\bar{r}_{ij,t}^{\text{com}}$ represent the minimum and maximum compression ratios of compressor (i, j) , respectively.

The gas turbine serves as a coupling unit between the GDN and PDN, and is given by

$$P_{i,t}^{\text{gt}} = \eta_i^{\text{gt}} \omega_i^{\text{ng}} f_{i,t}^{\text{gt}}, \quad \forall i \in \mathcal{V}^{g,\text{gt}}, t \in [0, T], \quad (9)$$

where $P_{i,t}^{\text{gt}}$ is the power output of the gas turbine at node i and time t , $f_{i,t}^{\text{gt}}$ is the corresponding natural gas consumption, and η_i^{gt} is the turbine efficiency.

Gas sources continuously provide a stable gas supply and pressure to the natural gas network. The pressure at a source node is assumed to be fixed. Accordingly, the gas supply and pressure constraints of the gas sources are written as

$$\underline{f}_i^{\text{well}} \leq f_{i,t}^{\text{well}} \leq \bar{f}_i^{\text{well}}, \quad \forall i \in \mathcal{V}^{g,\text{well}}, t \in [0, T], \quad (10a)$$

$$p_{i,t} = \bar{p}_i^{\text{well}}, \quad \forall i \in \mathcal{V}^{g,\text{well}}, t \in [0, T], \quad (10b)$$

where $f_{i,t}^{\text{well}}$ denotes the natural gas flow supplied by gas source i at time t , $\underline{f}_i^{\text{well}}$ and $\bar{f}_i^{\text{well}}$ are its lower and upper bounds, respectively, and $\bar{p}_i^{\text{well}}$ is the fixed pressure of gas source i .

The nodal gas flow balance and pressure continuity constraints of the GDN can be expressed as

$$\sum_{(j,i) \in \mathcal{E}^{g,\text{pip}}} f_{ji,t}(L_{ji},t) + \sum_{(j,i) \in \mathcal{E}^{g,\text{com}}} f_{ji,t}^{\text{com,out}} + f_{i,t}^{\text{well}} + \sum_{k \in \mathcal{K}} \lambda_{k,i}^{\text{h,gn}} \sigma_{kv} f_{k,t}^s = \sum_{(i,j) \in \mathcal{E}^{g,\text{pip}}} f_{ij}(0,t) + \sum_{(i,j) \in \mathcal{E}^{g,\text{com}}} f_{ij,t}^{\text{com,in}} + f_{i,t}^{\text{gt}} + f_{i,t}^{\text{ld}} - f_{i,t}^{\text{sh}}, \quad \forall i \in \mathcal{V}^g, t \in [0, T], \quad (11a)$$

$$\begin{cases} p_{i,t} = p_{ij}(0,t), & \forall (i,j) \in \mathcal{E}^{g,\text{pip}}, \\ p_{i,t} = p_{ji}(L_{ji},t), & \forall (j,i) \in \mathcal{E}^{g,\text{pip}}, \end{cases} \quad \forall i \in \mathcal{V}^g, t \in [0, T], \quad (11b)$$

where $f_{i,t}^{\text{ld}}$ denotes the gas load at node i at time t ; $f_{i,t}^{\text{sh}}$ denotes the amount of gas load shedding at node i at time t ; σ_{kv} is the dimension conversion factor; and $\lambda_{k,i}^{\text{h,gn}} \in \{0, 1\}$ is the connection-status parameter between the HESS located at k and node i of the GDN. Specifically, $\lambda_{k,i}^{\text{h,gn}} = 1$ if the HESS at k is connected to node i , and $\lambda_{k,i}^{\text{h,gn}} = 0$ otherwise.

In the literature on dynamic optimization of WDNs, water flow in pipeline $(i, j) \in \mathcal{E}^{w,\text{pip}}$ is generally governed by the water hammer equations [37], which constitute a system of HPDEs:

$$\frac{\partial h_{ij}}{\partial t} + \frac{a^2}{g A_{ij}} \frac{\partial q_{ij}}{\partial x} = 0, \quad (12a)$$

$$\frac{\partial q_{ij}}{\partial t} + g A_{ij} \left(\frac{\partial h_{ij}}{\partial x} + \frac{dh_{ij}^{\text{ele}}}{dx} \right) + \frac{\lambda q_{ij}^2}{2D_{ij} A_{ij}} = 0, \quad (12b)$$

where h_{ij} and q_{ij} denote the pressure head and volumetric flow rate in pipeline (i, j) , respectively, which are functions of space $x \in [0, L_{ij}]$ and time $t \in [0, T]$; a denotes the pressure wave speed; h_{ij}^{ele} represents the elevation head along the pipeline (i, j) ; and g denotes the gravitational acceleration.

To describe the hydraulic head relationships between different types of edges and nodes, the nodal head variable $\hat{h}_{i,t}$ is introduced to represent the total hydraulic head at node $i \in \mathcal{V}^w$ during the time period t . The total head consists of two components, namely the pressure head $h_{i,t}$ and the elevation head $h_{i,t}^{\text{ele}}$, such that

$$\hat{h}_{i,t} = h_{i,t} + h_{i,t}^{\text{ele}}. \quad (13)$$

In this paper, the pressure head is treated as a state variable, while the elevation head is considered a known constant.

For commonly used constant-speed centrifugal pumps, both the hydraulic head gain and the power consumption are functions of the flow rate. For each pump $(i, j) \in \mathcal{E}^{w,\text{pump}}$, its mathematical formulations are given by Guo et al. [38]:

$$\Delta h_{ij,t}^p = \hat{h}_{j,t} - \hat{h}_{i,t} = (\omega_{ij,t}^p)^2 \left[\bar{h}_{ij}^p - r_{ij}^p \left(\frac{q_{ij,t}^p}{\omega_{ij,t}^p} \right)^{n_{ij}^p} \right], \quad (14a)$$

$$P_{ij,t}^p = \frac{\rho^w g \Delta h_{ij,t}^p q_{ij,t}^p}{1000 \eta_{ij}^p}, \quad (14b)$$

$$Q_{ij,t}^p = P_{ij,t}^p \tan(\arccos(p_{ij}^p)), \quad (14c)$$

where $q_{ij,t}^p$ denotes the water flow rate through pump (i, j) during time period t ; $P_{ij,t}^p$ and $Q_{ij,t}^p$ represent the active and reactive power consumption of the pump, respectively; $\Delta h_{ij,t}^p$ denotes the hydraulic head gain provided by the pump; $\bar{h}_{ij}^p$, r_{ij}^p , and n_{ij}^p are the static head coefficient, internal resistance coefficient, and head exponent of the pump, respectively, which can be obtained by fitting the pump characteristic curves;

p_{ij}^p denotes the power factor of the pump; $\omega_{ij,t}^p$ represents the relative rotational speed of the pump, defined as the ratio of the operating speed to the rated speed, and is assumed to be unity for constant-speed pumps; ρ^w denotes the water density; and η_{ij}^p is the pump efficiency.

Reservoirs in the WDN are assumed to have sufficiently large capacities, and the pressure head at reservoir nodes is considered to be zero. Accordingly, the reservoir model can be expressed as

$$q_{i,t}^{\text{rer}} \geq 0, \quad \forall i \in \mathcal{V}^{\text{w.rer}}, t \in [0, T], \quad (15a)$$

$$q_{i,t}^{\text{rer}} = \sum_{(i,j) \in \mathcal{E}^{\text{w.pip}}} q_{ij}(0, t) + \sum_{(i,j) \in \mathcal{E}^{\text{w.pump}}} q_{ij,t}^p, \quad \forall i \in \mathcal{V}^{\text{w.rer}}, t \in [0, T], \quad (15b)$$

$$\hat{h}_{i,t} = h_{i,t}^{\text{ele}}, \quad \forall i \in \mathcal{V}^{\text{w.rer}}, t \in [0, T], \quad (15c)$$

where $q_{i,t}^{\text{rer}}$ denotes the outflow of the reservoir at node i at time t .

Storage tanks serve as water storage components in the WDN. The relationships among the tank pressure head, storage volume, and inlet/outlet flows are given by

$$h_{i,t+1} = h_{i,t} + \frac{\Delta t}{\pi(D_i^{\text{tank}}/2)^2} (q_{i,t}^{\text{tank,in}} - q_{i,t}^{\text{tank,out}}), \quad \forall i \in \mathcal{V}^{\text{w.tank}}, t \in [0, T], \quad (16a)$$

$$V_{i,t+1}^{\text{tank}} = V_{i,t}^{\text{tank}} + (q_{i,t}^{\text{tank,in}} - q_{i,t}^{\text{tank,out}}) \Delta t, \quad \forall i \in \mathcal{V}^{\text{w.tank}}, t \in [0, T], \quad (16b)$$

$$q_{i,t}^{\text{tank,in}} = \sum_{(j,i) \in \mathcal{E}^{\text{w.pip}}} q_{ji}(L_{ji}, t) + \sum_{(j,i) \in \mathcal{E}^{\text{w.pump}}} q_{ji,t}^p, \quad \forall i \in \mathcal{V}^{\text{w.tank}}, t \in [0, T], \quad (16c)$$

$$q_{i,t}^{\text{tank,out}} = \sum_{(i,j) \in \mathcal{E}^{\text{w.pip}}} q_{ij}(0, t) + \sum_{(i,j) \in \mathcal{E}^{\text{w.pump}}} q_{ij,t}^p, \quad \forall i \in \mathcal{V}^{\text{w.tank}}, t \in [0, T], \quad (16d)$$

where $V_{i,t}^{\text{tank}}$ denotes the water volume stored in the tank at node i at time t , $q_{i,t}^{\text{tank,in}}$ and $q_{i,t}^{\text{tank,out}}$ denote the inlet and outlet flows of the tank at node i at time t , respectively, and D_i^{tank} is the inner diameter of the tank at node i .

The nodal mass balance and head continuity constraints of the WDN can be written as

$$\begin{aligned} \sum_{(j,i) \in \mathcal{E}^{\text{w.pip}}} q_{ji}(L_{ji}, t) + \sum_{(j,i) \in \mathcal{E}^{\text{w.pump}}} q_{ji,t}^p &= \sum_{(i,j) \in \mathcal{E}^{\text{w.pip}}} q_{ij}(0, t) + \sum_{(i,j) \in \mathcal{E}^{\text{w.pump}}} q_{ij,t}^p \\ &+ q_{i,t}^{\text{ld}} - q_{i,t}^{\text{sh}} + \sum_{k \in \mathcal{K}} \lambda_{k,i}^{\text{h.wn}} q_{k,t}^{\text{elz}}, \quad \forall i \in \mathcal{V}^w \setminus (\mathcal{V}^{\text{w.rer}} \cup \mathcal{V}^{\text{w.tank}}), t \in [0, T], \end{aligned} \quad (17a)$$

$$\begin{cases} \hat{h}_{i,t} = \hat{h}_{ij}(0, t), & \forall (i, j) \in \mathcal{E}^{\text{w.pip}}, \\ \hat{h}_{i,t} = \hat{h}_{ji}(L_{ji}, t), & \forall (j, i) \in \mathcal{E}^{\text{w.pip}}, \end{cases} \quad \forall i \in \mathcal{V}^w, t \in [0, T], \quad (17b)$$

where $q_{i,t}^{\text{ld}}$ denotes the water demand at node i at time t , $q_{i,t}^{\text{sh}}$ denotes the amount of water load shedding at node i at time t , $\lambda_{k,i}^{\text{h.wn}} \in \{0, 1\}$ is the connection status parameter between the HESS at location k and node i of the WDN. Specifically, $\lambda_{k,i}^{\text{h.wn}} = 1$ if the HESS at location k is connected to node i , and $\lambda_{k,i}^{\text{h.wn}} = 0$ otherwise.

A linearized branch flow model is adopted to represent the AC power flow in a radial PDN [39]. Constraint (18a) ensures that if line (i, j) is closed, one bus is the parent of the other. Constraints (18b) and (18c) establish the relationship between line status and outages. Constraints (18d) and (18e) enforce active and reactive power balance at each bus, respectively. Additional linear constraints, such as radial structure limits and voltage limits, can be found in [40].

$$\alpha_{ij,t} = \beta_{ij,t} + \beta_{ji,t}, \quad \forall (i, j) \in \mathcal{E}^e, t \in \mathcal{T}, \quad (18a)$$

$$\alpha_{ij,t} \leq \vartheta_{ij,t}, \quad \forall (i, j) \in \mathcal{E}_{\text{rcs}}^e, t \in \mathcal{T}, \quad (18b)$$

$$\alpha_{ij,t} = \vartheta_{ij,t} \tilde{\alpha}_{ij}, \quad \forall (i, j) \in \mathcal{E}^e \setminus \mathcal{E}_{\text{rcs}}^e, t \in \mathcal{T}, \quad (18c)$$

$$\begin{aligned} \sum_{(j,i) \in \mathcal{E}^e} P_{ji,t} + P_{i,t}^{\text{sub}} + \sum_{k \in \mathcal{K}} \lambda_{k,i}^{\text{h,en}} P_{k,t}^{\text{s,e}} + P_{[i]^g,t}^{\text{gt}} + P_{i,t}^{\text{reg}} &= \\ \sum_{(i,j) \in \mathcal{E}^e} P_{ij,t} + P_{[i]^w,t}^p + P_{i,t}^{\text{ld}} - P_{i,t}^{\text{sh}} + \sum_{k \in \mathcal{K}} \lambda_{k,i}^{\text{h,en}} P_{k,t}^{\text{b,e}}, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \end{aligned} \quad (18d)$$

$$\begin{aligned} \sum_{(j,i) \in \mathcal{E}^e} Q_{ji,t} + Q_{i,t}^{\text{sub}} + Q_{[i]^g,t}^{\text{gt}} + Q_{i,t}^{\text{reg}} &= \\ \sum_{(i,j) \in \mathcal{E}^e} Q_{ij,t} + Q_{[i]^w,t}^p + Q_{i,t}^{\text{ld}} - Q_{i,t}^{\text{sh}}, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \end{aligned} \quad (18e)$$

where $\alpha_{ij,t}$ is a binary variable indicating the status of line (i, j) , taking the value of one if the line is closed and zero otherwise; $\beta_{ij,t}$ is a binary variable that equals one if bus j is selected as the parent of bus i in the radial operating topology; $\vartheta_{ij,t}$ is a binary parameter indicating whether line (i, j) is available, taking the value of zero if the line is damaged and one otherwise; $\tilde{\alpha}_{ij}$ denotes the initial status of line (i, j) ; $P_{ij,t}$ and $Q_{ij,t}$ denote the active and reactive power flows on line (i, j) ; $P_{i,t}^{\text{sub}}$ and $Q_{i,t}^{\text{sub}}$ denote the active and reactive power imported from the substation at bus i ; $P_{[i]^g,t}^{\text{gt}}$ and $Q_{[i]^g,t}^{\text{gt}}$ denote the active and reactive power outputs of the gas turbine at bus i , and $[i]^g$ indicates the index of the associated gas turbine in the GDN; $P_{i,t}^{\text{reg}}$ and $Q_{i,t}^{\text{reg}}$ denote the active and reactive power outputs of REG at bus i ; $P_{i,t}^{\text{ld}}$ and $Q_{i,t}^{\text{ld}}$ denote the active and reactive loads at bus i ; $P_{i,t}^{\text{sh}}$ and $Q_{i,t}^{\text{sh}}$ denote the curtailed portions of the active and reactive loads at bus i ; $P_{[i]^w,t}^p$ and $Q_{[i]^w,t}^p$ denote the active and reactive power consumptions of the water pump electrically connected to bus i , and $[i]^w$ indicates the index of the associated pump in the WDN; and $\lambda_{k,i}^{\text{h,en}} \in \{0, 1\}$ is a connection parameter indicating whether the HESS at candidate site k is connected to bus i .

3.3. Optimal HESS planning problem formulation

The objective is to minimize the total planning cost, as formulated in (19), which consists of the investment cost Ω^{inv} and the expected operating cost $\mathbb{E}[\Omega^{\text{ope}}]$. The investment cost accounts for the fixed cost of HESS siting and the capacity-dependent costs of internal components. Let $\mathcal{A} := \{\text{elz, ht, fc, mr, hrs}\}$ denote the set of HESS components, where elz, ht, fc, mr, and hrs represent the electrolyzer, hydrogen tank, solid oxide fuel cell, methanation reactor, and hydrogen refueling station, respectively. The operating cost includes load shedding costs in the PDN, WDN, and GDN, along with a penalty for unserved hydrogen refueling demand.

$$\min \Omega^{\text{inv}} + \mathbb{E}[\Omega^{\text{ope}}], \quad (19a)$$

$$\Omega^{\text{inv}} = \sum_{k \in \mathcal{K}} (c^{\text{hess}} \delta_k + \sum_{* \in \mathcal{A}} c^* C_k^*), \quad (19b)$$

$$\Omega^{\text{ope}} = N^d \sum_{t \in \mathcal{T}} \left(\sum_{i \in \mathcal{V}^e} p^e P_{i,t}^{\text{sh}} + \sum_{i \in \mathcal{V}^g} p^g f_{i,t}^{\text{sh}} + \sum_{i \in \mathcal{V}^w} p^w q_{i,t}^{\text{sh}} + \sum_{k \in \mathcal{K}} p^d q_{k,t}^{\text{sh}} \right) \Delta t, \quad (19c)$$

where δ_k denotes a binary variable that equals 1 if a HESS is deployed at site k ; C_k^* and c^* denote the installed capacity and annual investment cost of component $* \in \mathcal{A}$ at site k , respectively; c^{hess} denotes the annual investment cost of HESS deployment; p^e , p^g , and p^w are the unit costs of electricity, gas, and water load shedding, respectively; p^d denotes the unit penalty cost for unserved hydrogen refueling demand; $P_{i,t}^{\text{sh}}$, $f_{i,t}^{\text{sh}}$, and $q_{i,t}^{\text{sh}}$ denote the amounts of electricity, gas, and water load shedding at node i and time t ; and N^d is a time-scaling factor used to extrapolate the expected annual cost to the total expected cost over the entire study horizon.

To control the overall investment scale, constraints are imposed on the number of deployed HESSs as well as the capacity configurations of their internal components, which can be expressed as follows:

$$\sum_{k \in \mathcal{K}} \delta_k = N, \quad (20a)$$

$$0 \leq C_k^* \leq \delta_k \bar{C}^*, \quad \forall * \in \mathcal{A}, k \in \mathcal{K}, \quad (20b)$$

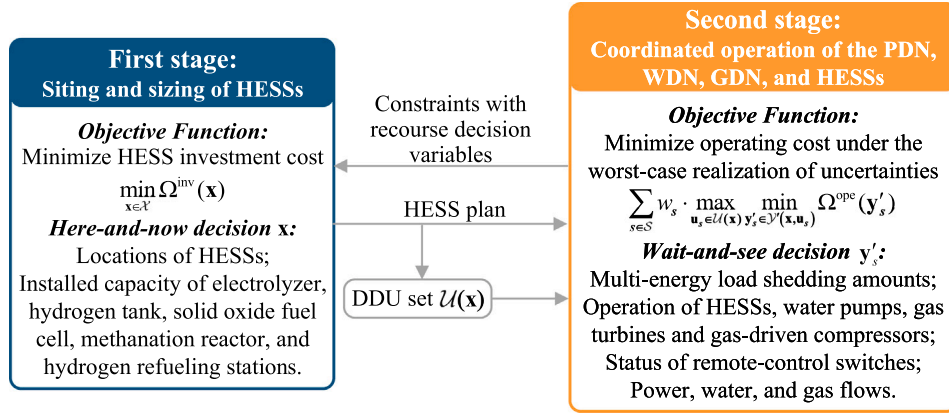


Fig. 2. Two-Stage Stochastic-Robust Optimization Strategy for HESS Planning.

where N denotes the total number of HESSs to be deployed, and δ_k is a binary decision variable indicating whether a HESS is installed at candidate site k . The parameters $\bar{C}^{\text{elz}}$, $\bar{C}^{\text{ht}}$, $\bar{C}^{\text{fc}}$, $\bar{C}^{\text{mr}}$, and $\bar{C}^{\text{hrs}}$ denote the upper bounds on the installed capacities of the electrolyzer, hydrogen tank, solid oxide fuel cell, methanation reactor, and hydrogen refueling station, respectively.

4. Two-stage stochastic-robust optimization strategy

The proposed strategy for handling endogenous and exogenous uncertainties in HESS planning is illustrated in Fig. 2. In this strategy, the uncertainties in $P_{i,t}^{\text{reg}}$, $P_{i,t}^{\text{ld}}$, $f_{i,t}^{\text{ld}}$, $q_{i,t}^{\text{ld}}$, and $\theta_{i,j,t}$ are modeled as exogenous DIUs and represented by a finite set of scenarios with assigned probabilities [41], since they are mainly driven by weather conditions, user consumption, and contingency events rather than by HESS investment decisions. In contrast, the hydrogen refueling demand $d_{k,t}$ is affected by HESS deployment, as the siting and sizing of HESSs change refueling accessibility, service capability, detour distance, and the perceived reliability of hydrogen supply. If $d_{k,t}$ were described by a conventional exogenous uncertainty set, the same demand uncertainty would be imposed on all HESS planning schemes, thereby ignoring the effect of HESS deployment on refueling demand. This may underestimate the induced refueling demand under deployment schemes with improved accessibility and service capability, overestimate the hydrogen available for multi-energy resilience support, and bias HESS siting, sizing, and operation decisions. Therefore, $d_{k,t}$ is modeled by a polyhedral continuous DDU set whose structure depends on the investment decisions of HESSs. Let $\mathbf{x}$ denote the first-stage investment decision vector, and $\mathbf{y}'$ denote the second-stage operational decision vector. Define the endogenous uncertainty vector as $\mathbf{u} := (d_{k,t})_{k \in \mathcal{K}, t \in \mathcal{T}}$ and $\mathcal{U}(\mathbf{x})$ as its DDU set. The investment constraints of the HESSs are collected to form the first-stage feasible region $\mathcal{X}$, and the operational constraints of the HESSs, PDN, WDN, and GDN are collected to form the second-stage feasible region $\mathcal{Y}'(\mathbf{x}, \mathbf{u})$, given $\mathbf{x} \in \mathcal{X}$ and a realization of endogenous uncertainty $\mathbf{u} \in \mathcal{U}(\mathbf{x})$. Let S denote the set of scenarios, with $s \in S$ being the scenario index, and each scenario s is associated with a probability w_s . Note that $\mathbf{y}'$, $\mathbf{u}$, $\mathcal{Y}'(\mathbf{x}, \mathbf{u})$, and Ω^{ope} under scenario s are denoted by $\mathbf{y}'_s$, $\mathbf{u}_s$, $\mathcal{Y}'_s(\mathbf{x}, \mathbf{u}_s)$, and Ω_s^{ope} respectively.

As shown in Fig. 2, the first-stage problem determines $\mathbf{x}$ prior to the realization of any uncertainty, aiming to minimize $\Omega^{\text{inv}}(\mathbf{x})$. The decision $\mathbf{x}$ is regarded as a *here-and-now* decision. The second stage is formulated as a scenario-based stochastic-robust optimization problem. For each scenario $s \in S$, a max-min optimization problem is solved to evaluate the worst-case operational performance. In each scenario s , the outer maximization identifies the worst-case realization of $\mathbf{u}_s$ within $\mathcal{U}(\mathbf{x})$, and the inner minimization determines $\mathbf{y}'_s \in \mathcal{Y}'_s$ to minimize Ω_s^{ope} , given $\mathbf{x}$ and $\mathbf{u}_s$. The decision $\mathbf{y}'_s$ is treated as a *wait-and-see* or recourse decision.

The expected operating cost is then computed by aggregating the worst-case costs across all scenarios, weighted by their respective probabilities. The scenario-wise constraints with recourse decision variables under the identified worst-case realization are then generated and added to the first-stage problem.

To couple HFCV refueling demand with the EWGN planning model without introducing a full transportation network assignment problem, the hydrogen refueling demand of HFCVs is aggregated at the PDN nodes $\mathcal{V}^e$. Here, a PDN node is interpreted as an aggregated energy-service zone or demand centroid, rather than a physical road intersection or a specific refueling origin. Specifically, let $d_{i,t}$ denote the uncertain hydrogen refueling demand associated with the service zone represented by PDN node i . To link the nodal hydrogen demand with the station hydrogen demand $d_{k,t}$, the following relationship is introduced:

$$\sum_{i \in \mathcal{V}^e} d_{i,t} = \sum_{k \in \mathcal{K}} d_{k,t}, \quad \forall t \in \mathcal{T}. \quad (21)$$

Specifically, each aggregated service zone is assumed to have a baseline level of hydrogen refueling demand. If an HESS is deployed in the vicinity of the service zone represented by PDN node $i \in \mathcal{V}^e$, additional induced hydrogen demand may arise because the deployment improves local refueling accessibility, service capability, and perceived hydrogen supply reliability. Let $\mathcal{K}_i \subseteq \mathcal{K}$ denote the set of candidate HESS locations that are geographically adjacent to the service zone represented by PDN node i . Based on these considerations, a DDU set $\mathcal{U}(\mathbf{x})$ is constructed to characterize the induced variations in HFCV hydrogen demand resulting from HESS siting and sizing decisions, as given by

$$\mathcal{U}(\mathbf{x}) = \left\{ \mathbf{u} \mid \begin{array}{l} (21) \\ d_{i,t} = d_{i,t}^{\text{base}} + d_{i,t}^{\text{ind}}, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \\ \underline{d}_{i,t}^{\text{base}} \leq d_{i,t}^{\text{base}} \leq \bar{d}_{i,t}^{\text{base}}, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \\ \sum_{k \in \mathcal{K}_i} \xi_{k,t} \delta_k \leq d_{i,t}^{\text{ind}} \leq \sum_{k \in \mathcal{K}_i} \bar{\xi}_{k,t} \delta_k, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \\ \left| d_{i,t}^{\text{ind}} - \frac{1}{2} \sum_{k \in \mathcal{K}_i} \delta_k (\xi_{k,t} + \bar{\xi}_{k,t}) \right| \leq \bar{d}_{i,t}, \quad \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \\ \sum_{i \in \mathcal{V}^e} \bar{d}_{i,t} \leq \tau \frac{\sum_{k \in \mathcal{K}} C_k^{\text{hrs}}}{\sum_{i \in \mathcal{V}^e} \bar{d}_{i,t}^{\text{base}}}, \quad \forall t \in \mathcal{T}, \\ \underline{d}_t + \kappa_t \sum_{k \in \mathcal{K}} C_k^{\text{hrs}} \leq \sum_{i \in \mathcal{V}^e} d_{i,t} \leq \bar{d}_t + \bar{\kappa}_t \sum_{k \in \mathcal{K}} C_k^{\text{hrs}}, \quad \forall t \in \mathcal{T}. \end{array} \right. \quad (22)$$

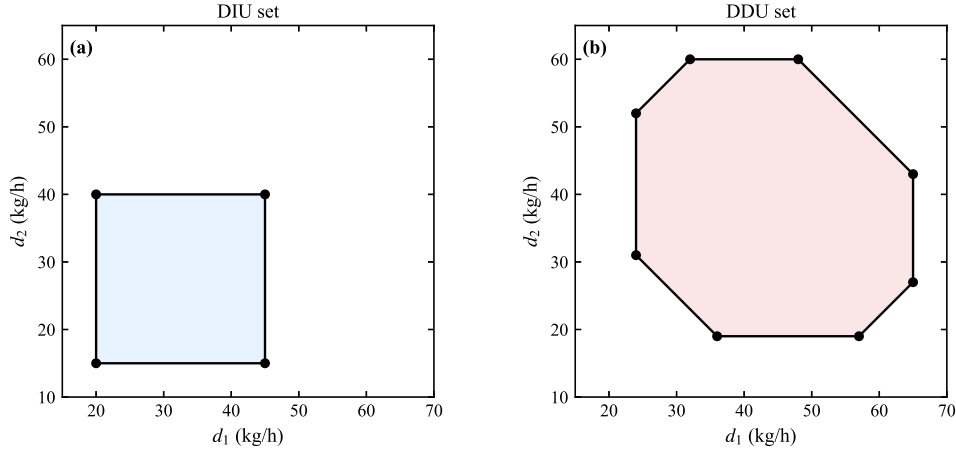


Fig. 3. Illustrative Comparison of the DIU and DDU sets for HFCV hydrogen refueling demand: (a) DIU set; (b) DDU set.

where $d_{i,t}^{\text{base}}$ and $d_{i,t}^{\text{ind}}$ denote the baseline and induced components of the hydrogen refueling demand at node i and time t , respectively; $\underline{d}_{i,t}^{\text{base}}$ and $\bar{d}_{i,t}^{\text{base}}$ denote the lower and upper bounds of the baseline hydrogen demand at node i , respectively; $\xi_{k,t}$ and $\bar{\xi}_{k,t}$ represent the lower and upper induction coefficients associated with the HESS located at candidate site k , respectively; δ_k is a binary variable indicating whether a HESS is deployed at site k , and thus determines whether the corresponding induced demand can occur; $\bar{d}_{i,t}$ denotes a disturbance term capturing deviations of the induced demand from its nominal value; $\underline{d}_t$ and $\bar{d}_t$ denote the lower and upper bounds of the aggregate hydrogen demand at time t , respectively; and τ , κ_t , and $\bar{\kappa}_t$ are scaling coefficients that control the budgets of local disturbances and global demand deviations. These parameters in the DDU set $\mathcal{U}(\mathbf{x})$ can be calibrated from empirical data or specified based on planning assumptions and scenario projections, depending on data availability. Specifically, $\underline{d}_{i,t}^{\text{base}}$ and $\bar{d}_{i,t}^{\text{base}}$ can be estimated from historical or forecasted nodal HFCV refueling profiles; $\underline{d}_t$ and $\bar{d}_t$ can be obtained from system-level HFCV hydrogen demand forecasts or by aggregating nodal demand profiles with appropriate uncertainty margins; $\xi_{k,t}$ and $\bar{\xi}_{k,t}$ can be calibrated using station-level refueling data, service coverage, accessibility improvement, or scenario-based demand projections; κ_t and $\bar{\kappa}_t$ can be determined according to the expected system-level demand response to the deployed hydrogen refueling station capacity; and τ can be selected according to the desired robustness level.

The first constraint decomposes the hydrogen demand at each node into a baseline component and an induced component. The second constraint specifies the uncertainty bounds of the baseline hydrogen demand. The third constraint indicates that, once the HESS siting decisions are determined, the induced demand at each node is obtained as a weighted aggregation of the induction coefficients associated with nearby HESS locations, thereby explicitly capturing the impact of HESS deployment on demand induction. The fourth constraint introduces a disturbance term to model deviations of the induced demand from its nominal value, allowing for both upward and downward fluctuations. The fifth constraint imposes a budget on the aggregate disturbance level, such that the sum of nodal deviations is limited by a ratio determined by the total baseline demand and the aggregated hydrogen refueling station capacity, thereby preventing excessive overall deviations. The sixth constraint enforces a global bound on the total hydrogen demand at each time, ensuring that the aggregate demand remains within an acceptable range determined by the system configuration.

When the induction coefficients $\xi_{k,t}$ and $\bar{\xi}_{k,t}$, as well as the scaling parameters τ , κ_t , and $\bar{\kappa}_t$, are all set to zero, the DDU set $\mathcal{U}(\mathbf{x})$ degenerates

into a conventional DIU set $\tilde{\mathcal{U}}(\mathbf{x})$, as follows:

$$\tilde{\mathcal{U}}(\mathbf{x}) = \left\{ \mathbf{u} \mid \begin{cases} \underline{d}_{i,t}^{\text{base}} \leq d_{i,t} \leq \bar{d}_{i,t}^{\text{base}}, & \forall i \in \mathcal{V}^e, t \in \mathcal{T}, \\ \underline{d}_t \leq \sum_{i \in \mathcal{V}^e} d_{i,t} \leq \bar{d}_t, & \forall t \in \mathcal{T}. \end{cases} \right. \quad (23)$$

This observation indicates that the DIU set $\tilde{\mathcal{U}}(\mathbf{x})$ is a special case of the proposed DDU set $\mathcal{U}(\mathbf{x})$ when the hydrogen refueling demand exhibits no dependence on HESS deployment decisions.

Fig. 3 provides an illustrative example of the conventional DIU set and the proposed DDU set for two aggregated HFCV hydrogen refueling demand zones. The illustrative two-dimensional uncertainty sets are generated using the following parameters. The baseline demand bounds of the two HFCV refueling demand zones are set as $d_1^{\text{base}} \in [20, 45]$ kg/h and $d_2^{\text{base}} \in [15, 40]$ kg/h. In the DIU case, the aggregate demand bounds are set as $30 \leq d_1 + d_2 \leq 90$. In the DDU case, one HESS is assumed to be deployed with $\delta_1 = 1$ and $C_1^{\text{hrs}} = 100$, affecting both demand zones with $\mathcal{K}_1 = \mathcal{K}_2 = \{1\}$. The induction coefficients are set as $\xi_1 = 4$ kg/h and $\bar{\xi}_1 = 20$ kg/h, and the budget parameter is set as $\tau = 6.8$. The coefficients are set as $\kappa = 0.25$ and $\bar{\kappa} = 0.18$, leading to the DDU aggregate bounds $55 \leq d_1 + d_2 \leq 108$. As shown in Fig. 3(a), the DIU set treats hydrogen refueling demand as an exogenous uncertainty independent of the HESS deployment decision. Therefore, the feasible demand region remains a fixed rectangle determined by the baseline demand bounds. In contrast, Fig. 3(b) shows that the DDU set is expanded and reshaped after the HESS is deployed. This is because the DDU set explicitly accounts for the demand induction effect through the investment-dependent induction terms, the uncertainty budget, and the capacity-dependent aggregate demand bounds. Hence, compared with the DIU set, the DDU set captures the fact that HESS planning decisions may change the uncertainty region of future hydrogen refueling demand. If this effect is ignored, the resulting HESS planning solution may underestimate the induced refueling demand and lead to insufficient hydrogen supply capability in practical operation.

Based on the proposed strategy, the developed HPDE-constrained HESS planning model can be formulated as the following two-stage stochastic-robust optimization problem with DDU:

$$\min_{\mathbf{x} \in \mathcal{X}} \Omega^{\text{inv}}(\mathbf{x}) + \sum_{s \in \mathcal{S}} w_s \max_{\mathbf{u}_s \in \mathcal{U}(\mathbf{x})} \min_{\mathbf{y}'_s \in \mathcal{Y}'_s(\mathbf{x}, \mathbf{u}_s)} \Omega^{\text{ope}}(\mathbf{y}'_s). \quad (24)$$

5. PC&CG-based tailored solution algorithm

The optimization problem in (24) is computationally intractable due to the HPDE constraints (7) and (12), as well as the nonlinear constraints (14a) and (14b). To address this challenge, this section develops a tailored solution algorithm that reformulates (24) into a computationally tractable MILP by combining numerical discretization and piecewise linear relaxation. Specifically, numerical discretization is first applied to convert HPDE constraints (7) and (12) into finite-dimensional algebraic constraints. This step transforms the HPDE-constrained network dynamics into an algebraic form, but the resulting formulation still contains nonlinear algebraic terms. Piecewise linear relaxation is then applied to these remaining nonlinearities, including the nonlinear flow and pump relations, to obtain MILP-compatible linear constraints. After these two reformulation steps, the original HPDE-constrained optimization problem is reformulated as a two-stage stochastic-robust MILP problem, which is solved iteratively using a modified PC&CG algorithm.

The Preissmann four-point implicit finite difference scheme, also known as the box scheme, is adopted to discretize HPDEs (7) and (12) in space and time [36,37]. For each pipeline $(i, j) \in \mathcal{E}^{\text{S}, \text{pip}} \cup \mathcal{E}^{\text{W}, \text{pip}}$, the continuous space-time domain $[0, L_{ij}] \times [0, T]$ is divided into a mesh of discrete points spaced with space step Δx_{ij} and time step Δt . Let $\mathcal{T}^+ := \{0\} \cup \mathcal{T}$ denote the set of time indices, and let $\mathcal{N}_{ij}^+ := \{0, 1, \dots, N_{ij}\}$ denote the set of space indices along pipeline (i, j) . Each variable or parameter defined at grid point $(n, m) \in \mathcal{N}_{ij}^+ \times \mathcal{T}^+$ is indexed by subscript n and superscript m . The discretization of (7) yields:

$$A_{ij} \frac{p_{n+1,ij}^{m+1} + p_{n,ij}^{m+1} - p_{n+1,ij}^m - p_{n,ij}^m}{2\Delta t} + a^2 \frac{f_{n+1,ij}^{m+1} - f_{n,ij}^{m+1}}{\Delta x_{ij}} = 0, \quad (25a)$$

$$\frac{f_{n+1,ij}^{m+1} + f_{n,ij}^{m+1} - f_{n+1,ij}^m - f_{n,ij}^m}{2\Delta t} + A_{ij} \frac{p_{n+1,ij}^{m+1} - p_{n,ij}^{m+1}}{\Delta x_{ij}} + \frac{a^2}{A_{ij}} \frac{\phi_{n+1,ij}^{m+1} - \phi_{n,ij}^{m+1}}{\Delta x_{ij}} = -\frac{\lambda a^2}{4D_{ij}A_{ij}}(\phi_{n+1,ij}^{m+1} + \phi_{n,ij}^{m+1}), \quad (25b)$$

where auxiliary variable $\phi_{n,ij}^m$ is defined as

$$\phi_{n,ij}^m = (f_{n,ij}^m)^2 / r_{n,ij}^m. \quad (26)$$

The discretization of (12) yields:

$$\frac{h_{n+1,ij}^{m+1} + h_{n,ij}^{m+1} - h_{n+1,ij}^m - h_{n,ij}^m}{2\Delta t} + \frac{a^2}{gA_{ij}} \frac{q_{n+1,ij}^{m+1} - q_{n,ij}^{m+1}}{\Delta x_{ij}} = 0, \quad (27a)$$

$$\frac{q_{n+1,ij}^{m+1} + q_{n,ij}^{m+1} - q_{n+1,ij}^m - q_{n,ij}^m}{2\Delta t} + gA_{ij} \frac{h_{n+1,ij}^{m+1} - h_{n,ij}^{m+1}}{\Delta x_{ij}} + \frac{\lambda}{4D_{ij}A_{ij}}(\psi_{n+1,ij}^{m+1} + \psi_{n,ij}^{m+1}) = 0, \quad (27b)$$

where auxiliary variable $\psi_{n,ij}^m$ is defined as

$$\psi_{n,ij}^m = (q_{n,ij}^m)^2. \quad (28)$$

Through numerical discretization, HPDE constraints (7) and (12) are transformed into linear algebraic constraints (25) and (27), as well as nonlinear nonconvex algebraic constraints (26) and (28) for computational solvability.

Nonlinear nonconvex constraints (26), (28), (14a) and (14b) still lead to high computational complexity. Specifically, (28) is a quadratic equality constraint; (14a) indicates that the pump head gain $\Delta h_{ij,t}^p$ is a univariate nonlinear function of the pump flow rate $q_{ij,t}^p$; and (14b) contains a bilinear term $\Delta h_{ij,t}^p q_{ij,t}^p$. By introducing an auxiliary variable $\zeta_{n,ij}^m$, (26) can be equivalently rewritten as:

$$\zeta_{n,ij}^m = (f_{n,ij}^m)^2, \quad (29a)$$

$$\zeta_{n,ij}^m = \phi_{n,ij}^m p_{n,ij}^m. \quad (29b)$$

To address these nonlinearities, piecewise linear relaxation techniques are employed [42]. Consider a monotonic function $f(x)$ over

$\underline{x} \leq x \leq \bar{x}$. Its domain $[\underline{x}, \bar{x}]$ is divided into M partitions, and the partition set is $\{(x_{m-1}, x_m) : m = 1, \dots, M\}$, where (x_{m-1}, x_m) is a pair of values that denote the lower and upper bounds of x in partition m . A continuous variable $b'_m \in [0, 1]$ is assigned to x_m and a binary variable b''_m is assigned to partition m . A piecewise linear relaxation of the graph of $f(x)$ is given by

$$\langle f(x) \rangle_I := \left\{ \begin{array}{l} (x, y) \in \mathbb{R}^2 \mid b'_0 \leq b''_1, b'_M \leq b''_M \\ b'_m \leq b''_m + b''_{m+1}, m = 1, \dots, M-1 \\ x = \sum_{m=0}^M b'_m x_m, y \leq \sum_{m=0}^M b'_m f(x_m) \\ \sum_{m=0}^M b'_m = 1, \sum_{m=1}^M b''_m = 1 \\ y \geq f'(x_m)(x - x_m) + f(x_m), m = 0, \dots, M. \end{array} \right\}.$$

Then, (14a), (28) and (29a) can be replaced by

$$\langle q_{ij,t}^p, \Delta h_{ij,t}^p \rangle \in \langle \Delta h_{ij,t}^p(q_{ij,t}^p) \rangle_I \quad (30a)$$

$$\langle q_{n,ij}^m, \psi_{n,ij}^m \rangle \in \langle \psi_{n,ij}^m(q_{n,ij}^m) \rangle_I \quad (30b)$$

$$\langle f_{n,ij}^m, \zeta_{n,ij}^m \rangle \in \langle \zeta_{n,ij}^m(f_{n,ij}^m) \rangle_I. \quad (30c)$$

Consider a bilinear function $x \cdot y$ over $\underline{x} \leq x \leq \bar{x}$ and $\underline{y} \leq y \leq \bar{y}$. The domain $[\underline{y}, \bar{y}]$ of y is divided into I partitions, and the partition set is $\{(\underline{y}_i, \bar{y}_i) : i = 1, \dots, I\}$, where $(\underline{y}_i, \bar{y}_i)$ is a pair of values that denote the lower and upper bounds of y in partition i . A binary variable a_i and two continuous variables x_i, y_i are assigned to partition i . A piecewise linear relaxation of the graph of $x \cdot y$ was proposed in [43] and is defined as

$$\langle x \cdot y \rangle_{II} := \left\{ \begin{array}{l} (x, y, z) \in \mathbb{R}^3 \mid \sum_{i=1}^I a_i = 1, y = \sum_{i=1}^I y_i, x = \sum_{i=1}^I x_i \\ z \geq \sum_{i=1}^I (\underline{x} y_i + \underline{y}_i x_i - \underline{x} \underline{y}_i a_i), z \geq \sum_{i=1}^I (\bar{x} y_i + \bar{y}_i x_i - \bar{x} \bar{y}_i a_i) \\ z \leq \sum_{i=1}^I (\bar{x} y_i + \underline{y}_i x_i - \bar{x} \underline{y}_i a_i), z \leq \sum_{i=1}^I (\underline{x} y_i + \bar{y}_i x_i - \underline{x} \bar{y}_i a_i) \\ \underline{y}_i a_i \leq y_i \leq \bar{y}_i a_i, \underline{x}_i a_i \leq x_i \leq \bar{x}_i a_i, i = 1, \dots, I. \end{array} \right\}.$$

Then, (29b) and (14b) can be replaced by

$$p_{ij,t}^p = \frac{\rho^w g Z_{ij,t}^p}{1000 \eta_{ij}^p} \quad (31a)$$

$$\langle \Delta h_{ij,t}^p, q_{ij,t}^p, Z_{ij,t}^p \rangle \in \langle \Delta h_{ij,t}^p \cdot q_{ij,t}^p \rangle_{II}, \quad (31b)$$

$$\langle \phi_{n,ij}^m, p_{n,ij}^m, \zeta_{n,ij}^m \rangle \in \langle \phi_{n,ij}^m \cdot p_{n,ij}^m \rangle_{II}. \quad (31c)$$

Let $\mathbf{y}_s$ and $\mathcal{Y}_s(\mathbf{x}, \mathbf{u}_s)$ denote, respectively, the decision variable vector and feasible region of the second-stage problem under scenario s after numerical discretization and piecewise linear relaxations. Then, the HPDE-constrained HESS planning model can be reformulated as an MILP problem, and its compact form is given by:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{c}_1 \mathbf{x} + \sum_{s \in \mathcal{S}} \omega_s \max_{\mathbf{u}_s \in \mathcal{U}_s(\mathbf{x})} \min_{\mathbf{y}_s \in \mathcal{Y}_s(\mathbf{x}, \mathbf{u}_s)} \mathbf{c}_2 \mathbf{y}_s, \quad (32)$$

where

$$\mathcal{X} = \{\mathbf{x} \in \{0, 1\}^{b_x} \times \mathbb{R}_+^{c_x} : \mathbf{A} \mathbf{x} \geq \mathbf{b}\}, \quad (33a)$$

$$\mathcal{Y}_s(\mathbf{x}, \mathbf{u}_s) = \{\mathbf{y}_s \in \{0, 1\}^{b_y} \times \mathbb{R}_+^{c_y} : \mathbf{B}_s \mathbf{y}_s \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s\}, \quad (33b)$$

$$\mathcal{U}(\mathbf{x}) = \{\mathbf{u}_s \in \mathbb{R}_+^{c_u} : \mathbf{F}(\mathbf{x}) \mathbf{u}_s \leq \mathbf{h} + \mathbf{G} \mathbf{x}\}. \quad (33c)$$

Here, $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{b}$, $\mathbf{h}$, and $\mathbf{d}_s$ are coefficient vectors, and $\mathbf{A}$, $\mathbf{B}_s$, $\mathbf{H}_s$, $\mathbf{E}_s$, $\mathbf{F}(\mathbf{x})$, and $\mathbf{G}$ denote constraint matrices. In (32), $\mathbf{x}$ includes the HESS siting and

sizing decisions; under scenario s , $\mathbf{y}_s$ collects variables related to HESS operation, PDN reconfiguration and dispatch, discretized WDN/GDN flow operation, load shedding, and auxiliary variables introduced by piecewise linear relaxation; and $\mathbf{u}_s$ collects the decision-dependent HFCV hydrogen refueling demand variables.

The finite difference scheme and piecewise linear relaxation may introduce approximation errors. However, these errors can be effectively controlled in practical planning applications. The error associated with the finite difference scheme can be reduced by refining the temporal and spatial discretization grids, while the approximation error caused by piecewise linear relaxation decreases as the partition size becomes smaller. In addition, optimization-based bound tightening can be performed prior to piecewise linear relaxation to narrow the feasible domains of the relevant variables, thereby improving the tightness of the piecewise linear relaxation without introducing an excessive number of partitions [42]. Before practical implementation, the obtained HESS planning solution can be further validated using the original nonlinear equations. If the resulting violations exceed the prescribed tolerances, the discretization grids or partitions can be further refined. Therefore, the approximation errors can be effectively controlled within acceptable limits, reducing the risk of infeasible or unsafe operation in practical HESS planning.

After numerical discretization and piecewise linear relaxation, the proposed HESS planning model in (32) is reformulated as a two-stage stochastic-robust MILP problem with mixed-integer recourse and DDU. This problem structure is not directly suitable for standard Benders decomposition. Classical Benders decomposition relies on dualizing the recourse subproblem to generate valid cuts for the master problem, whereas the proposed model contains mixed-integer recourse, a max-min robust subproblem, and an uncertainty set that changes with the first-stage HESS investment decision. Traditional C&CG is also not directly applicable, since it usually assumes a fixed uncertainty set. In contrast, the worst-case hydrogen refueling demand in this work depends on $\mathbf{x}$. Hence, uncertainty realizations generated under one investment decision may not remain valid or representative under another. Therefore, this study adopts the PC&CG algorithm proposed in [26], with necessary modifications to handle multiple stochastic scenarios in the second stage. PC&CG generates parametric columns and constraints to capture the dependence between the uncertainty realization and the first-stage decision, making it suitable for the stochastic-robust MILP problem (32) with DDU. Only the essential solution steps are outlined here, and please refer to [26] for details of the original PC&CG algorithm.

Let $\mathbf{y}_s^c$ and $\mathbf{y}_s^d$ denote the continuous and discrete components of $\mathbf{y}_s$, i.e., $\mathbf{y}_s = (\mathbf{y}_s^c, \mathbf{y}_s^d)$. By relaxing $\mathbf{y}_s^d$ to continuous variables in $[0, 1]$, the following equivalence holds:

$$\max_{\mathbf{u}_s \in \mathcal{U}_s(\mathbf{x})} \min\{\mathbf{c}_2 \mathbf{y}_s : \mathbf{y}_s \in [0, 1]^{b_y} \times \mathbb{R}_+^{c_y}, \mathbf{B}_s \mathbf{y}_s \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s\} \quad (34a)$$

$$= \max\{(\mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s)^\top \boldsymbol{\pi}_s : \mathbf{u}_s \in \mathcal{U}(\mathbf{x}), \mathbf{B}_s^\top \boldsymbol{\pi}_s \leq \mathbf{c}_2^\top, \boldsymbol{\pi}_s \geq 0\}. \quad (34b)$$

where $\boldsymbol{\pi}_s$ are the dual variables of the constraints in (34a).

Let $\Pi_s = \{\boldsymbol{\pi}_s : \mathbf{B}_s^\top \boldsymbol{\pi}_s \leq \mathbf{c}_2^\top, \boldsymbol{\pi}_s \geq 0\}$. Denote by $\mathcal{P}_s$ and $\mathcal{R}_s$ the sets of extreme points and extreme rays of Π_s , respectively. Let $\mathcal{Y}_s^r(\mathbf{x}^*, \mathbf{u})$ denote the linear programming relaxation of $\mathcal{Y}_s(\mathbf{x}^*, \mathbf{u}_s)$. Define the problem $\mathbf{LP}_s(\mathbf{x}, \boldsymbol{\beta}) : \max\{(-\mathbf{E}_s \mathbf{u}_s)^\top \boldsymbol{\beta} : \mathbf{u}_s \in \mathcal{U}(\mathbf{x})\}$. When $\boldsymbol{\beta} = \boldsymbol{\pi}_s \in \mathcal{P}_s$, let $\mathcal{OU}_s(\mathbf{x}, \boldsymbol{\pi}_s)$ denote the optimal solution set of $\mathbf{LP}_s(\mathbf{x}, \boldsymbol{\pi}_s)$, which is given by (35). Similarly, when $\boldsymbol{\beta} = \boldsymbol{\gamma}_s \in \mathcal{R}_s$, let $\mathcal{OV}_s(\mathbf{x}, \boldsymbol{\gamma}_s)$ denote the optimal solution set of $\mathbf{LP}_s(\mathbf{x}, \boldsymbol{\gamma}_s)$. This can be obtained by replacing $\mathbf{u}_s$ with $\mathbf{v}_s$ and λ_s with ζ_s in (35), where ζ_s denotes the corresponding dual variables.

$$\mathcal{OU}_s(\mathbf{x}, \boldsymbol{\pi}_s) = \left\{ \begin{array}{l} \mathbf{F}(\mathbf{x})\mathbf{u}_s \leq \mathbf{h} + \mathbf{G}\mathbf{x} \\ \mathbf{F}(\mathbf{x})^\top \lambda_s \geq -\mathbf{E}_s^\top \boldsymbol{\pi}_s \\ \lambda_s \circ (\mathbf{h} + \mathbf{G}\mathbf{x} - \mathbf{F}(\mathbf{x})\mathbf{u}_s) = 0 \\ \mathbf{u}_s \circ (\mathbf{F}(\mathbf{x})^\top \lambda_s + \mathbf{E}_s^\top \boldsymbol{\pi}_s) = 0 \\ \mathbf{u}_s \geq 0, \quad \lambda_s \geq 0 \end{array} \right\} \quad (35)$$

where λ_s denotes the dual variables of constraints in $\mathcal{U}(\mathbf{x})$.

For each scenario $s \in \mathcal{S}$, let $\mathbf{y}_s^{\pi_s}$ and $\mathbf{u}_s^{\pi_s}$ denote the second-stage decision and uncertainty realization corresponding to the extreme point $\boldsymbol{\pi}_s \in \mathcal{P}_s$. Similarly, let $\mathbf{y}_s^{\gamma_s}$ and $\mathbf{v}_s^{\gamma_s}$ denote the corresponding variables associated with the extreme ray $\boldsymbol{\gamma}_s \in \mathcal{R}_s$. Then, the MILP problem in (32) is decomposed into an investment master problem **MP** and multiple scenario-specific operation subproblems **SP_s**, as follows:

$$\mathbf{MP} : \underline{\rho} = \min \mathbf{c}_1 \mathbf{x} + \sum_{s \in \mathcal{S}} w_s \eta_s \quad (36a)$$

$$\text{s.t.} \quad \mathbf{A}\mathbf{x} \geq \mathbf{b} \quad (36b)$$

$$\eta_s \geq \mathbf{c}_2^\top \mathbf{y}_s^{\pi_s} \quad \forall \boldsymbol{\pi}_s \in \mathcal{P}_s, s \in \mathcal{S} \quad (36c)$$

$$\mathbf{B}_s \mathbf{y}_s^{\pi_s} \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s^{\pi_s} \quad \forall \boldsymbol{\pi}_s \in \mathcal{P}_s, s \in \mathcal{S} \quad (36d)$$

$$(\mathbf{u}_s^{\pi_s}, \lambda_s^{\pi_s}) \in \mathcal{OU}_s(\mathbf{x}, \boldsymbol{\pi}_s) \quad \forall \boldsymbol{\pi}_s \in \mathcal{P}_s, s \in \mathcal{S} \quad (36e)$$

$$\mathbf{B}_s \mathbf{y}_s^{\gamma_s} \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{v}_s^{\gamma_s} \quad \forall \boldsymbol{\gamma}_s \in \mathcal{R}_s, s \in \mathcal{S} \quad (36f)$$

$$(\mathbf{v}_s^{\gamma_s}, \zeta_s^{\gamma_s}) \in \mathcal{OV}_s(\mathbf{x}, \boldsymbol{\gamma}_s) \quad \forall \boldsymbol{\gamma}_s \in \mathcal{R}_s, s \in \mathcal{S} \quad (36g)$$

$$\mathbf{x} \in \{0, 1\}^{b_x} \times \mathbb{R}_+^{c_x}, \quad \mathbf{y}_s^{\gamma_s}, \mathbf{y}_s^{\pi_s} \in \{0, 1\}^{b_y} \times \mathbb{R}_+^{c_y} \quad (36h)$$

$$\mathbf{SP}_s : \max_{\mathbf{u}_s \in \mathcal{U}(\mathbf{x}^*)} \min\{\mathbf{c}_2 \mathbf{y}_s : \mathbf{y}_s \in \mathcal{Y}_s^r(\mathbf{x}^*, \mathbf{u}_s)\}. \quad (37)$$

Moreover, define three additional subproblems for a given $\mathbf{x}^*$ and s :

$$\mathbf{SP1}_s : \eta_s^f(\mathbf{x}^*) = \max_{\mathbf{u}_s \in \mathcal{U}(\mathbf{x}^*)} \min\{\mathbf{1}^\top \tilde{\mathbf{y}}_s : \mathbf{y}_s \in \{0, 1\}^{b_y} \times \mathbb{R}_+^{c_y}, \tilde{\mathbf{y}}_s \geq 0, \mathbf{B}_s \mathbf{y}_s + \tilde{\mathbf{y}}_s \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x}^* - \mathbf{E}_s \mathbf{u}_s\}, \quad (38)$$

$$\mathbf{SP2}_s : \eta_s^s(\mathbf{x}^*, \mathbf{y}_s^{d*}) = \max_{\mathbf{u}_s \in \mathcal{U}(\mathbf{x}^*)} \min\{\mathbf{c}_2 \mathbf{y}_s : \mathbf{y}_s \in \mathcal{Y}_s(\mathbf{x}^*, \mathbf{u}_s), \mathbf{y}_s^d = \mathbf{y}_s^{d*}\}, \quad (39)$$

$$\mathbf{SP3}_s : \max\{(\mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s^f)^\top \boldsymbol{\pi}_s : \boldsymbol{\pi}_s \in \Pi_s\}. \quad (40)$$

Based on these definitions, Algorithm 1 outlines the tailored solution algorithm for the HPDE-constrained HESS planning. To address complementarity constraints arising in the solution of **MP**, **SP_s**, **SP1_s**, **SP2_s** and **SP3_s**, the big-M approach is employed to obtain equivalent linear formulations [26]. It is worth noting that, in each iteration, the scenario-specific subproblems are solved independently in Lines 11–19 of Algorithm 1, thereby enabling parallel computation.

6. Case studies

6.1. Simulation system and parameter settings

To validate the effectiveness and superiority of the proposed method for the EWGN, a benchmark coupled system is constructed by combining a modified IEEE 33-bus PDN [44], the Belgium 20-node GDN [45], and the Richmond 40-node WDN [46]. The overall topology of the coupled system is illustrated in Fig. 4.

In the PDN, four gas turbines (GT1–GT4) are installed with maximum active power outputs of 1 MW, 1 MW, 2 MW, and 1 MW, respectively. In the GDN, gas sources GW1 and GW2 are connected to nodes 1 and 16, respectively, with fixed supply pressures of 4 bar and 3.9 bar. Two compressors, denoted as C1 and C2, are installed on pipelines (3,4) and (14,13), with compression ratios ranging from 1 to 1.8. To facilitate modeling and simulation, virtual nodes 21 and 22, as well as 23 and 24, are introduced at the inlet and outlet of compressors C1 and C2, respectively, thereby decomposing pipelines (3,4) and (14,13) into (3,21), (22,4), (14,23), and (24,13). The pressure limits of all gas pipelines are set within the range of [2, 4] bar. The main technical parameters of the gas pipelines are summarized in Table 1. In the WDN, a reservoir R1 supplies water to all network nodes. Seven pumps (P1–P7) are installed to provide hydraulic head support, and four storage tanks (S1–S4) are deployed. The allowable pressure head range of the water pipelines is [0, 30] m, and the key technical parameters are listed in Table 2. Based on historical operational data from an integrated energy system in a region of Hunan Province, China, probability density functions are fitted for renewable power outputs, electrical loads, water demands, and gas demands. Subsequently, Latin

Algorithm 1: PC&CG-based tailored solution algorithm.

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1 Reformulation: Obtain problem (32) by applying numerical
  discretization and piecewise linear relaxations to problem (24).
2 Set tolerance  $\varepsilon$ ,  $LB = -\infty$ ,  $UB = +\infty$ ,  $\mathcal{P}_s = \mathcal{R}_s = \emptyset \forall s \in S$ .
3 while  $UB - LB > \varepsilon$  do
4   Solve problem MP.
5   if problem MP is infeasible then
6     Report infeasibility of problem (32) and terminate.
7   else
8     Derive its optimal solution  $\mathbf{x}^*$  and optimal value  $\underline{o}$ .
9     Update  $LB = \underline{o}$ .
10  end
11  foreach  $s \in S$  do
12    Solve SP1s and derive its optimal solution of  $\mathbf{u}_s$ , denoted by  $\mathbf{u}_s^{f*}$ ,
    and optimal value  $\eta_s^f(\mathbf{x}^*)$ .
13    if  $\eta_s^f(\mathbf{x}^*) = 0$  then
14      Solve SPs to obtain optimal  $\mathbf{u}_s^*$  and extreme point  $\pi_s^*$  of  $\Pi_s$ ;
      Update  $\mathcal{P}_s = \mathcal{P}_s \cup \{\pi_s^*\}$ ; Create variables  $\mathbf{u}_s^{\pi_s^*}$  and  $\lambda_s^{\pi_s^*}$  and
      add (41) to MP.
      
$$\mathbf{y}_s^{\pi_s^*} \in \{0, 1\}^{b_s} \times \mathbb{R}_+^{c_s}, \eta_s \geq \mathbf{c}_2^\top \mathbf{y}_s^{\pi_s^*} \quad (41a)$$


$$\mathbf{B}_s \mathbf{y}_s^{\pi_s^*} \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{u}_s^{\pi_s^*} \quad (41b)$$


$$(\mathbf{u}_s^{\pi_s^*}, \lambda_s^{\pi_s^*}) \in \mathcal{O}U_s(\mathbf{x}, \pi_s^*). \quad (41c)$$

15    else
16      Solve SP3s to derive an extreme ray  $\gamma_s^*$  of  $\Pi_s$ ; Set
       $\eta_s^s(\mathbf{x}^*) = +\infty$ ; Update  $\mathcal{R}_s = \mathcal{R}_s \cup \{\gamma_s^*\}$ ; Create variables  $\mathbf{v}_s^{\gamma_s^*}$ 
      and  $\zeta_s^{\gamma_s^*}$  and add (42) to MP.
      
$$\mathbf{y}_s^{\gamma_s^*} \in \{0, 1\}^{b_s} \times \mathbb{R}_+^{c_s}, \quad (42a)$$


$$\mathbf{B}_s \mathbf{y}_s^{\gamma_s^*} \geq \mathbf{d}_s - \mathbf{H}_s \mathbf{x} - \mathbf{E}_s \mathbf{v}_s^{\gamma_s^*} \quad (42b)$$


$$(\mathbf{v}_s^{\gamma_s^*}, \zeta_s^{\gamma_s^*}) \in \mathcal{O}V_s(\mathbf{x}, \gamma_s^*). \quad (42c)$$

17    end
18    Solve the original MILP recourse problem in (32) for given  $\mathbf{x}^*$ 
    and  $\mathbf{u}_s^*$  to obtain optimal  $(\mathbf{y}_s^c, \mathbf{y}_s^d)$ ; Solve SP2s with given  $\mathbf{x}^*$ 
    and  $\mathbf{y}_s^d$  to obtain  $\eta_s^s(\mathbf{x}^*, \mathbf{y}_s^d)$ .
19  end
20  Update  $UB = \min\{UB, \mathbf{c}_1^\top \mathbf{x}^* + \sum_{s \in S} w_s \eta_s^s(\mathbf{x}^*, \mathbf{y}_s^d)\}$ .
21 end
22 Output: The optimal HESS plan  $\mathbf{x}^*$ .

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hypercube sampling and K-means clustering methods proposed in [47] are employed to generate multiple representative operating scenarios, capturing the uncertainties of loads and renewable generation under different operating conditions. To further characterize power line outage risks, outage probabilities are calculated according to the annual average failure and repair rates of distribution lines. A Monte Carlo simulation is then conducted to randomly generate a large number of line operation and outage state combinations, which are further clustered to extract representative line outage scenarios along with their occurrence probabilities. Finally, the power line outage scenarios are combined with the electricity–gas–water load and renewable generation scenarios to construct three representative scenarios. For each scenario, the second-stage problem is defined over a 24-hour horizon with $\Delta t = 1$ h. Twelve candidate locations are considered for HESS deployment, and the number of deployed HESSs is set to three as a planning-scale parameter reflecting a limited deployment budget. The key parameters of the DDU are listed in Table 3. The spatial influence ranges of induced hydrogen demand and the coupling relationships between candidate HESS locations and the PDN, WDN, and GDN are provided in Table 4. The unit investment costs of the electrolyzer, hydrogen tank, methanation reactor, solid oxide fuel cell, and hydrogen refueling station are 292.34 CNY/kW/yr, 133.86 CNY/m³/yr, 7121.62 CNY/(m³/h)/yr, 24,925.61

CNY/kW/yr, and 24,759.29 CNY/(m³/h)/yr, respectively. The corresponding upper bounds of installed capacities are 3000 kW, 2000 m³, 200 m³/h, 2000 kW, and 300 m³/h. Hydrogen refueling station and hydrogen tank capacities are reported in volumetric units and have been converted to mass-based values for consistency in the simulation. For the HPDE discretization, each gas and water pipeline is uniformly divided into five spatial segments. This setting provides a balance between the representation of pipeline dynamics and the computational burden of the resulting MILP problem. The nonlinear terms are approximated by dividing the domains into five partitions, and optimization-based bound tightening is applied before piecewise linear relaxation [42]. The convergence tolerance of the PC&CG algorithm is set to $\varepsilon = 0.1\%$. All models and simulations are implemented on a laptop equipped with an Intel® Core™ i9-14900HX processor and 32 GB DDR5 memory. The simulation environment is based on the PyCharm platform. The optimization models are coded in Python 3.12 and solved using the solver Gurobi 13.0.

6.2. Simulation results and comparative analysis

Fig. 5 illustrates the convergence behavior of the proposed solution algorithm. The gap between the upper bound (UB) and the lower bound (LB) decreases monotonically as the number of iterations increases. After five iterations, the relative gap falls below the prescribed convergence tolerance, indicating that the algorithm converges successfully. This result verifies the effectiveness of the proposed solution algorithm in solving the HPDE-constrained decision-dependent robust planning problem for HESSs. The scalability of the proposed solution algorithm with respect to the number of scenarios is further examined. Table 5 reports the computational performance of the proposed solution algorithm under different numbers of scenarios. The results show that, as the number of scenarios increases from 1 to 20, the proposed solution algorithm consistently converges within six iterations. Moreover, the relative optimality gap for all test cases remains below the predefined threshold of 0.10%. These results demonstrate that the proposed solution algorithm is capable of efficiently solving the HPDE-constrained decision-dependent robust planning problem of HESSs, exhibiting both satisfactory computational efficiency and scalability.

Table 6 presents the planning results of the HESSs. According to (19a)–(19c), the planning model minimizes the sum of HESS investment cost and the expected operating cost associated with electricity, gas, and water load shedding as well as unmet HFCV refueling demand. Therefore, each HESS component is expanded only when its marginal contribution to reducing the operating penalty under outage scenarios is sufficient to compensate for its additional investment cost. The hydrogen storage capacities of HESS1–HESS3 all reach their upper bounds, indicating that sufficient hydrogen inventory has high marginal value for reallocating hydrogen among methanation, electricity support, and refueling service during line outages. In addition, the installed capacities of electrolyzers at all selected sites exceed 70% of their upper limits, reflecting the need to maintain adequate hydrogen production capability for multi-energy coordination and support. From a spatial perspective, the capacity differences across the selected HESSs are closely related to the local marginal value of each energy conversion pathway under the coupled network constraints. HESS3 is located at gas node 17 in the downstream region of the GDN and is close to node 19, where GT4 is connected. During line outages, GT4 tends to increase its power output to compensate for electricity shortages, which increases local natural gas withdrawal and tightens the gas pressure and flow constraints in the downstream GDN. Under the gas nodal balance and the gas flow constraints, synthetic natural gas injected by HESS3 directly alleviates downstream gas shortages and reduces gas load shedding. Since gas load shedding is penalized in (19c), methanation at HESS3 provides a high marginal operating-cost reduction. As a result, the optimization model assigns the largest methanation reactor capacity to HESS3. In contrast, HESS1 is located in the upstream region of the GDN, where

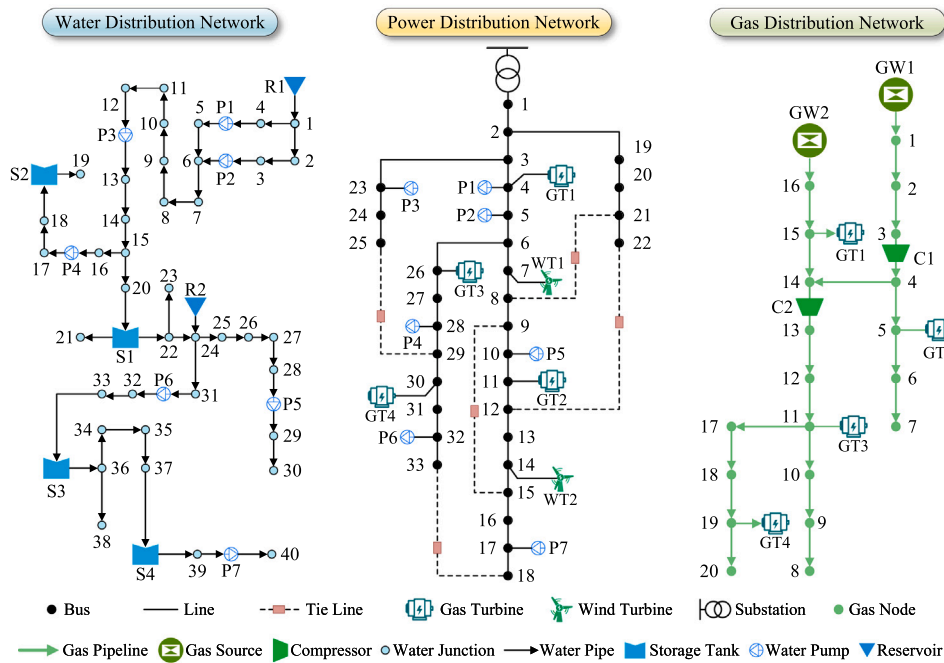


Fig. 4. Topology of the EWGN Test System.

gas supply conditions and pressure margins are relatively more favorable. Therefore, the marginal benefit of additional methanation capacity at HESS1 is lower than that at HESS3. The model consequently assigns a moderate methanation reactor capacity to HESS1 and reserves more conversion capability for electricity support through the solid oxide fuel cell, which helps reduce power load shedding and sustain electricity-driven water pump operation during line outages. HESS2 is deployed in the midstream region and serves a hydrogen refueling demand area with a relatively strong demand envelope. For this location, increasing hydrogen refueling station capacity is more effective in reducing the penalty of unmet HFCV demand, while the marginal benefit of additional methanation capacity is lower than that of HESS3 due to its different gas-network position. Therefore, HESS2 is equipped with the largest hydrogen refueling station capacity and the smallest methanation reactor capacity among the three selected HESSs.

The different solid oxide fuel cell and hydrogen refueling station capacities can also be explained by the hydrogen allocation constraints within each HESS. The electrolyzer and hydrogen tank provide a limited hydrogen resource that must be allocated among the methanation reactor, the solid oxide fuel cell, and the hydrogen refueling station. Hydrogen allocated to methanation reduces gas load shedding by injecting synthetic natural gas into the GDN, hydrogen allocated to the solid oxide fuel cell reduces electricity load shedding and supports electricity-driven water pumps, and hydrogen allocated to the refueling station reduces unmet HFCV demand. Therefore, the optimization model must balance these competing uses of hydrogen. The installed capacities of solid oxide fuel cells decrease from HESS1 to HESS3, indicating that HESS1 and HESS2 retain relatively strong electricity-support capability, whereas HESS3 allocates more of its hydrogen-related capacity to GDN support through methanation. Meanwhile, HESS2 is configured with the largest hydrogen refueling station capacity because reducing unmet HFCV demand at this location provides a larger operating-cost benefit than further increasing methanation capacity.

Overall, the planning results show that the selected HESS capacities are jointly driven by the coupled objective function, the network constraints, and the internal hydrogen allocation trade-offs. The model allocates larger methanation capacity to locations where synthetic natural gas injection has a higher marginal value for reducing gas load

Table 1

Key Technical parameters of gas Pipelines.

Pipeline	Length (m)	Diameter (m)	Pipeline	Length (m)	Diameter (m)
(1,2)	5000	0.3	(12,11)	5250	0.3
(2,3)	7500	0.3	(13,12)	5000	0.3
(3,4)	3250	0.3	(14,23)	3125	0.3
(4,21)	2687.5	0.2	(24,13)	3125	0.3
(22,5)	2687.5	0.2	(15,14)	1250	0.3
(5,6)	3625	0.2	(16,15)	3125	0.3
(6,7)	2375	0.2	(11,17)	1312.5	0.3
(4,14)	6875	0.3	(17,18)	3250	0.2
(9,8)	6250	0.2	(18,19)	12,250	0.2
(10,9)	2500	0.2	(19,20)	7500	0.2
(11,10)	3125	0.3	—	—	—

shedding, larger solid oxide fuel cell capacity to locations where electricity support is more valuable for maintaining PDN and WDN operation, and larger refueling station capacity to locations where reducing unmet HFCV demand provides a greater operating-cost benefit.

To further assess the contribution of HESSs to enhancing the resilience of the EWGN during line outages, a representative stochastic scenario is selected for detailed analysis. In this scenario, power lines (1,2), (6,7), (16,17), and (3,23) experience outages during 07:00–14:00, 07:00–18:00, 07:00–18:00, and 07:00–13:00, respectively. Fig. 6 presents the power outputs of GTs in the PDN, while Figs. 7–10 illustrate the operating results of methanation reactors, electrolyzers, hydrogen tanks, and solid oxide fuel cells within each HESS. During the outage of line (1,2), the PDN becomes islanded from the main grid, leading to a sharp reduction in available power supply. To avoid widespread power load shedding, GT2–GT4 operate at or near their maximum output levels for most of the outage period, whereas GT1 exhibits relatively lower output. GT3 and GT4 are located in downstream gas-network regions with higher local gas withdrawal during the outage period. To prevent local gas shortages in these regions, HESS3 provides the largest synthetic natural gas injection, while HESS2 provides a moderate but sustained gas injection during the critical outage hours. By contrast, HESS1 injects only a limited amount of synthetic natural gas,

Table 2
Key Technical parameters of water Pipelines.

Pipeline	Length (m)	Diameter (m)	Pipeline	Length (m)	Diameter (m)	Pipeline	Length (m)	Diameter (m)
(R1,1)	900	1	(S1,21)	660	0.8	(15,16)	700	0.8
(1,2)	600	1	(S1,22)	960	1	(17,18)	700	0.8
(1,4)	900	0.8	(22,23)	700	0.8	(18,S2)	700	0.8
(2,3)	500	1	(22,24)	700	0.8	(S2,19)	960	0.8
(5,6)	900	0.7	(24,25)	700	0.8	(15,20)	660	1
(6,7)	800	1	(25,26)	600	0.8	(20,S1)	960	0.8
(7,8)	800	1	(26,27)	700	0.8	(36,34)	400	0.8
(8,9)	800	1	(27,28)	700	0.8	(34,35)	400	0.7
(9,10)	800	1	(29,30)	300	0.8	(35,37)	384	0.7
(10,11)	700	1	(24,31)	400	0.8	(36,38)	400	0.7
(11,12)	700	1	(32,33)	400	0.8	(37,S4)	440	0.7
(13,14)	700	1	(33,S3)	500	0.8	(S4,39)	400	0.7
(14,15)	700	1	(S3,36)	384	0.8	–	–	–

Table 3
Key DDU Parameters.

Parameter	Time (h)					
	1–4	5–8	9–12	13–16	17–20	21–24
$d_{i,t}^{base}/\bar{d}_{i,t}^{base}$ (kg/h)	$1/\frac{5}{3}$	$4/2$	$5/17$	$5/17$	$4/2$	$1/\frac{5}{3}$
$\bar{z}_{k,t}/\bar{z}_{k,t}$ (kg/h)	$1/\frac{2}{3}$	$1/\frac{2}{3}$	$1/\frac{2}{3}$	$1/\frac{2}{3}$	$1/\frac{2}{3}$	$1/\frac{2}{3}$
$d_i/\bar{d}_i$ (kg/h)	$110/50$	$140/175$	$175/250$	$175/250$	$140/175$	$110/50$
$\kappa_i/\bar{\kappa}_i$	0.04/0.08	0.08/0.12	0.12/0.16	0.12/0.16	0.08/0.12	0.04/0.08

mainly outside the most critical gas-shortage interval, which is consistent with its lower marginal value for downstream gas support. As shown in Fig. 8, prior to the outage of line (1,2), electrolyzers intensively consume electricity and water to produce hydrogen, which is stored in hydrogen tanks in preparation for the forthcoming outage period. During the line outage, electrolyzers cease operation. Once line (1,2) is restored, hydrogen production resumes according to the remaining system requirements. This strategy of pre-charging during normal operation and discharging during outages enables HESSs to buffer energy deficits caused by power network disruptions. Fig. 9 shows the hydrogen tank operation under the representative scenario. Before the critical outage period, the three HESSs charge their hydrogen tanks to accumulate hydrogen inventory. During the outage period, the hydrogen storage levels decrease rapidly, indicating that stored hydrogen is released to support electricity generation, methanation, and hydrogen refueling services. After the main outage period, small charging and discharging actions appear according to the remaining system requirements. These results demonstrate that hydrogen tanks provide cross-period flexibility by shifting hydrogen availability from low-stress periods to critical periods with higher multi-energy support demand. As illustrated in Fig. 10, the solid oxide fuel cell in HESS2 provides the most sustained power output during the outage period, while HESS1 and HESS3 provide more intermittent peak-period support. This coordinated solid oxide fuel cell dispatch helps prevent power load shedding and supports the continuous operation of electricity-driven water pumps in the WDN.

Fig. 11 presents the simulated results of power, gas, and water load shedding, as well as unmet hydrogen refueling demand of HFCVs, under the representative scenario. As shown in the figure, no power load shedding occurs throughout the entire period. This indicates that, through the coordinated operation of GTs and the solid oxide fuel cells within HESSs, the PDN maintains a high level of supply reliability even during line outage conditions. In contrast, gas load shedding is mainly concentrated in the period from 07:00 to 14:00, reaching a peak value of approximately 93 m³/h around midday. During this interval, GTs operate at relatively high output levels to compensate for power shortages, which significantly increases natural gas consumption and intensifies operational pressure in the GDN. As a result, gas load shedding becomes unavoidable during certain periods. Water load shedding

Table 4
Induced hydrogen demand influence range and network coupling of HESSs.

Candidate Location	Affected Node Set	Connected PDN Node	Connected GDN Node	Connected WDN Node
1	{1, 2, 3, 4, 5, 6, 21, 22}	3	3	7
2	{3, 4, 5, 6, 7, 8, 9, 10}	6	5	9
3	{9, 10, 11, 12, 23, 24, 25}	10	6	13
4	{9, 10, 11, 12, 13, 14, 15}	13	12	15
5	{13, 14, 15, 16, 17, 18, 19, 20}	17	15	20
6	{5, 6, 7, 26, 27, 28}	6	14	22
7	{27, 28, 29, 30, 31, 32}	29	13	32
8	{28, 29, 30, 31, 32, 33}	31	17	24
9	{1, 2, 3, 4, 21, 22}	2	13	27
10	{10, 11, 12, 13, 14, 15, 16, 23}	12	12	34
11	{15, 16, 17, 18, 19, 20}	18	11	37
12	{6, 7, 8, 26, 27, 28, 29}	7	4	12

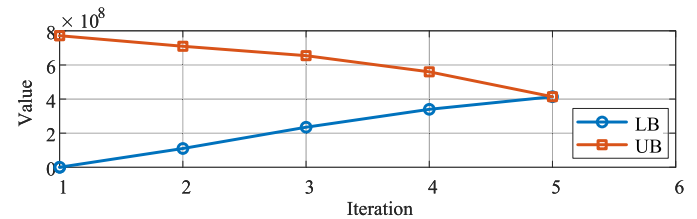


Fig. 5. Convergence behavior of the PC&CG-based tailored solution algorithm.

Table 5
Computational performance of the proposed solution method under different Numbers of Scenarios.

Number of Scenarios	Gap (%)	Solution time (h)	Number of Iterations
1	0.09	3.1	4
3	0.10	4.9	5
10	0.10	4.4	5
20	0.07	6.7	6

Table 6
HESS planning Results.

HESS	Location	Installed Capacity				
		Electrolyzer (kW)	Hydrogen Tank (m ³)	Methanation Reactor (m ³ /h)	Solid Oxide Fuel Cell(kW)	Hydrogen Refueling Station(m ³ /h)
1	1	2921.14	2000	65.33	1953.85	148.19
2	4	2233.13	2000	30.12	1784.12	205.42
3	8	2704.48	2000	80.76	1122.79	96.55

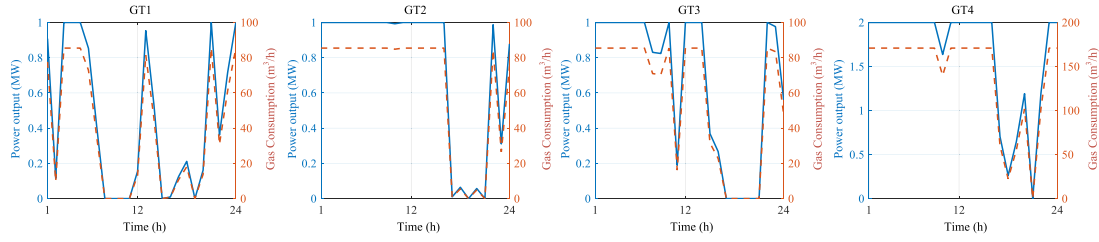


Fig. 6. GT power output results under a representative scenario.

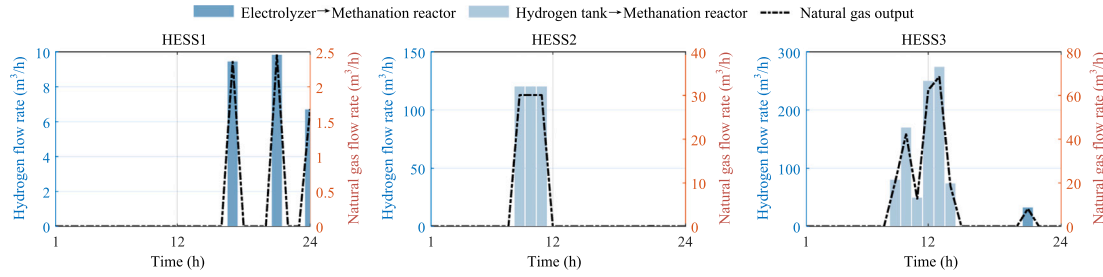


Fig. 7. Methanation reactor operating results under a representative scenario.

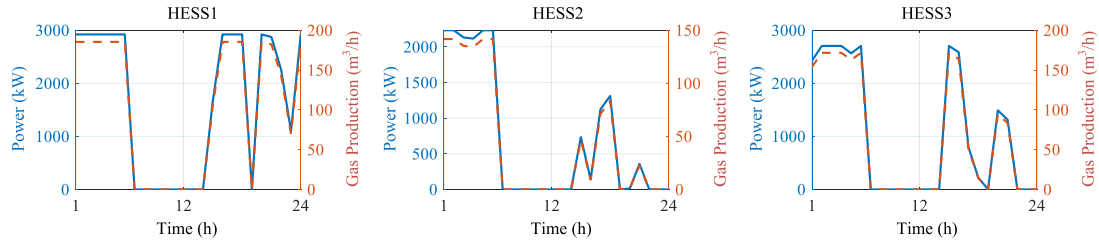


Fig. 8. Electrolyzer operating results under a representative scenario.

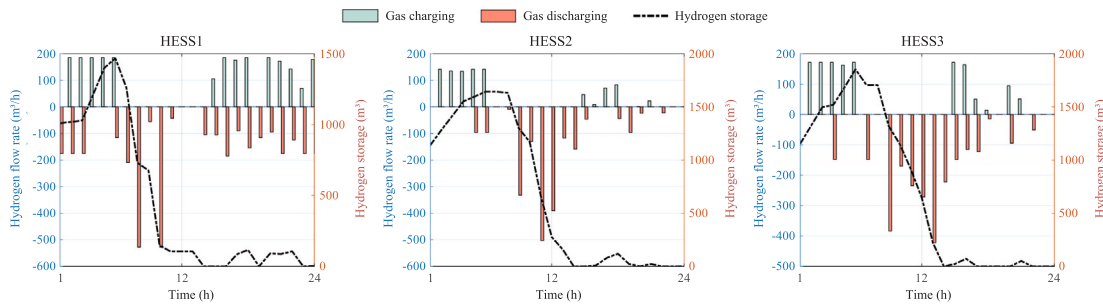


Fig. 9. Hydrogen tank operation results under a representative scenario.

remains limited in magnitude and exhibits intermittent fluctuations, mainly occurring in the morning and evening. This behavior is closely related to the operating status of electrolyzers. When electrolyzers are shut down in response to power and gas network conditions, water demand drops sharply; when hydrogen production resumes, water demand temporarily increases, leading to short-term variations in water load shedding. The unmet hydrogen refueling demand of HFCVs displays a pronounced peak-valley pattern, primarily concentrated between 10:00 and 12:00. This period coincides with the stage in which HESSs inject synthetic natural gas into the gas network via methanation. The results indicate that, under tight gas supply conditions, the system prioritizes maintaining gas network reliability by allocating hydrogen resources to

methanation, which constrains the hydrogen supply available for refueling stations and leads to partially unmet HFCV demand. The occurrence of unmet HFCV refueling demand during 10:00–12:00 is mainly caused by the hydrogen allocation trade-off within HESSs under gas-supply stress. During the outage period, GTs increase their power output to compensate for the reduced power supply from the PDN, which raises natural gas consumption and tightens the GDN pressure and flow constraints. In this case, allocating hydrogen to methanation can produce synthetic natural gas and reduce gas load shedding. However, the stored hydrogen in each HESS must be shared among methanation, solid oxide fuel cell generation, and HFCV refueling. Therefore, when the marginal benefit of reducing gas load shedding is higher, the optimization model

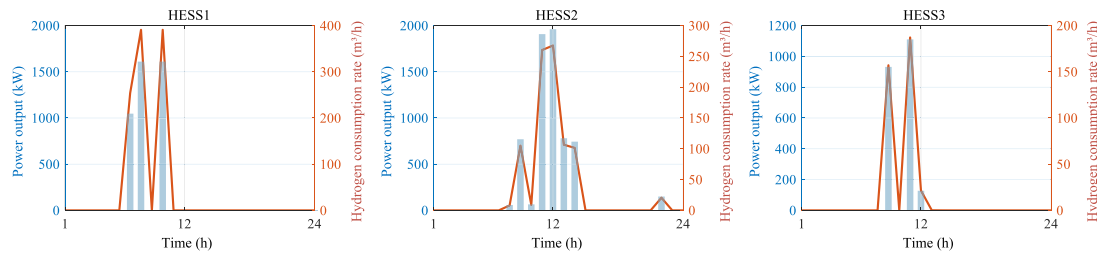


Fig. 10. Solid oxide fuel cell operation results under a representative scenario.

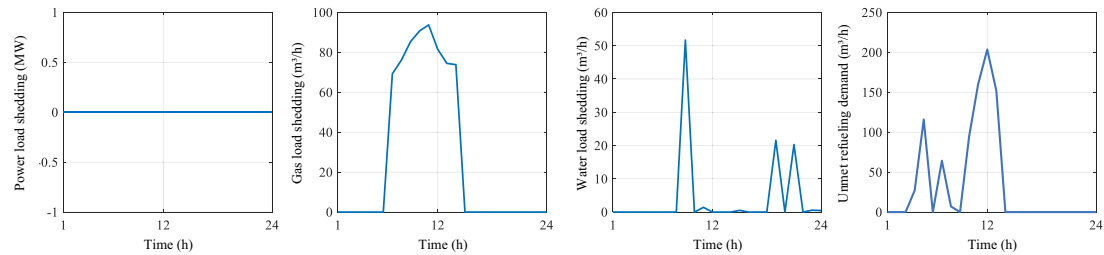


Fig. 11. Power, gas, and water load shedding and unmet HFCV refueling demand under a representative scenario.

Table 7
Performance Comparison of schemes 1–3.

Scheme	Power load shedding (MWh)	Water load shedding (m³)	Gas load shedding (m³)	Investment cost (10 ⁷ CNY)	Total load shedding cost (10 ³ CNY)	Investment return
1	0	96.49	645.15	1.366585	8.61	0.16
2	0.27	119.20	861.57	1.078364	9.21	0.12
3	5.53	310.01	1619.04	–	46.94	–

prioritizes methanation, which reduces the hydrogen available for refueling stations and results in partially unmet HFCV demand. Overall, by enabling flexible energy conversion and coordinated interactions among PDNs, WDNs and GDNs, HESSs allow the EWGN to prioritize electricity supply under line outages, while simultaneously mitigating gas load shedding and limiting unmet HFCV refueling demand. Meanwhile, the stable operation of water pumps is maintained, significantly reducing the risk of widespread multi-energy load shedding across the EWGN.

To evaluate the effectiveness and superiority of the proposed decision-dependent robust planning method for HESSs, three comparative schemes are designed in this paper: Scheme 1 applies the proposed method that explicitly accounts for DDU in HFCV hydrogen demand; Scheme 2 applies conventional planning methods that neglect the demand induction effect of HESS deployment and models HFCV hydrogen demand uncertainty using the DIU set $\hat{U}(x)$; Scheme 3 does not perform decision-dependent robust planning of HESSs, and the system performance of the EWGN is improved solely through coordinated multi-energy operation. Table 7 compares the simulation results of Schemes 1–3 under different performance metrics. In particular, the investment return on load shedding reduction is defined as the ratio of total investment cost to total load shedding cost. A higher value of this index indicates that a given level of investment yields a greater reduction in load shedding and thus reflects superior economic efficiency. Compared with Scheme 2, Scheme 1 incurs a 26.73% increase in investment cost but achieves a 6.51% reduction in total load shedding cost, resulting in a 33.33% improvement in the investment return. This demonstrates that explicitly considering DDU in HFCV hydrogen demand enables the proposed method to better cope with demand induction effects, thereby achieving a more favorable trade-off between investment cost

and load shedding cost. As a result, both economic efficiency and system resilience are improved. Compared with Scheme 3, Scheme 1 significantly reduces electricity, gas, and water load shedding. This result highlights that by leveraging multi-energy coupling and coordinated energy conversion among PDNs, WDNs, and GDNs enabled by HESSs, the overall resilience of the EWGN during line outages can be significantly enhanced.

7. Conclusion

This paper proposes a decision-dependent robust planning method for HESSs to enhance the resilience of the EWGN under exogenous and endogenous uncertainties. Multi-network synergies governed by HPDEs are exploited to mitigate multi-energy load shedding under line outages. The main findings are summarized as follows: 1) By enabling flexible multi-energy conversion among electricity, gas, and water through HESSs, the proposed method effectively mitigates the risk of widespread load shedding in the EWGN. Compared with the case without HESS planning, the total load shedding cost is reduced by 81.66%, highlighting the significant resilience benefits brought by HESS planning; 2) Compared with conventional planning methods that neglect the demand induction effect of HESS deployment, the proposed method reduces total load shedding cost by 6.51% while improving the investment return on load shedding reduction by 33.33%. This demonstrates that explicitly accounting for DDU in HFCV refueling hydrogen demand enables a more favorable balance between HESS investment cost and multi-energy load shedding, thereby achieving superior economic efficiency and system resilience for the EWGN. Future work will extend the proposed planning method to data-driven distributionally robust HFCV demand modeling and compound contingencies across the PDN, WDN, and GDN.

CRediT authorship contribution statement

Zhihao Hua: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yingping Cao:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Bin Zhou:** Writing – original draft, Resources, Methodology, Funding acquisition. **Siu Wing Or:** Writing – review & editing, Supervision, Funding acquisition. **Fan Yang:** Writing – original draft, Software, Data curation,

Conceptualization. **Canbing Li**: Writing – review & editing, Supervision, Resources, Methodology. **Ziqiang Wang**: Writing – review & editing, Writing – original draft, Software. **Yuexin Sun**: Writing – review & editing, Writing – original draft. **Xiangbin Yu**: Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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